Asymmetric Basis Momentum in Commodity Futures*

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Abstract

Basis momentum is implemented as a symmetric predictor: a move of the futures curve toward contango is treated as the mirror image of a move toward backwardation. In 43 commodity futures over 1960–2024, it is not. Positive basis momentum predicts next-month returns (slope 0.059, $t = 6.4$); negative basis momentum does not (0.002, $t = 0.2$), and the short positions of time-series basis momentum earned a Sharpe ratio of 0.06. The two sides carry different information. Only positive basis momentum forecasts the level of the futures curve, whereas the spread between the first two contracts is forecast mainly by negative basis momentum. Consistent with a convex scarcity rent, the positive-side slope is steeper in backwardation and when commercial hedgers are net short. A position with separate slopes for positive and negative signals earns an out-of-sample alpha of 0.30% per month beyond linear basis momentum and the commodity premium. The asymmetry is weak and insignificant after 2010.

Keywords: Basis Momentum, Commodity Futures, Convenience Yield, Hedging Pressure, Spot and Term Premiums, Return Predictability.

JEL codes: C45, G11, G12, G13, G17, Q02

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1 Introduction

Basis momentum, the difference between the past returns of the first- and the second-nearby futures contract on a commodity, is among the strongest predictors of commodity futures returns (Boons and Porras Prado, 2019). Like most commodity signals (Szymanowska, De Roon, Nijman, and Van Den Goorbergh, 2014; Koijen, Moskowitz, Pedersen, and Vrugt, 2018; Bakshi, Gao, and Rossi, 2019), it is implemented symmetrically, linearly or through ranks: a strongly negative value commands as large a short position as a strongly positive value commands a long one. This symmetry is an economic assumption, not a statistical convenience. It asserts that a futures curve moving toward contango carries the same information about future prices, with opposite sign, as a curve moving toward backwardation.

There are reasons to doubt it. The convenience yield of holding inventories is a convex function of inventories (Kaldor, 1939; Working, 1949; Brennan, 1958). When supply is ample, arbitrage pins the curve near full carry, so moves toward contango are bounded. When supply becomes scarce, the convenience yield rises steeply and the curve can move far into backwardation (Deaton and Laroque, 1992; Routledge, Seppi, and Spatt, 2000; Gorton, Hayashi, and Rouwenhorst, 2013). A move toward backwardation may therefore reveal a change in the risk premium that a move toward contango does not.

We find that the symmetry fails, and that it fails in a specific way: the relation is a hockey stick, flat for negative and rising for positive signals. In 43 commodity futures over 1960–2024, positive basis momentum predicts next-month risk-adjusted returns (slope 0.059, t-statistic of 6.4) and negative basis momentum does not (0.002, t-statistic of 0.2). The positions that time-series basis momentum takes on negative signals earned a Sharpe ratio of 0.06 over 1960–2024 and of -0.73 in 2000–2009, the decade in which linear basis momentum failed. Negative basis momentum is not empty, however. It predicts the return of the first- relative to the second-nearby contract at least as strongly as positive basis momentum does.

Basis momentum thus contains two kinds of information: one-sided information about the level of the futures curve, and information about the spread between the first two contracts, which comes mainly from the negative side. The standard implementation trades the spread information of the negative side as if it were information about the level.

The decomposition follows from a futures-return identity. The return of the nearby contract is the return of the second-nearby contract plus the spreading return between the two. Positive basis momentum predicts the second-nearby return as strongly as the nearby return, and negative basis momentum predicts neither. The spreading return is predicted mainly by negative basis momentum, and the slopes of the two sides do not differ significantly. This reconciles the hockey stick with the finding of Boons and Porras Prado (2019) that the spreading premium of basis momentum comes largely from its short leg, and with the profitability of the short leg of the cross-sectional basis-momentum factor. Negative basis momentum carries relative information. It says that the front of the curve will underperform the next contract, not that the commodity will fall.

The asymmetry is specific to basis momentum. Time-series momentum, the basis, the relative basis of Gu, Kang, Lou, and Tang (2025), and the hedging pressure of commercial traders are not asymmetric on their own, and none of them explains the asymmetry of basis momentum. Time-series momentum, for example, predicts returns equally for positive and negative signals. The long side of basis momentum therefore does not dominate merely because commodity futures earn a positive average return.

Scarcity is the interpretation most consistent with the state dependence of the positive side. The slope on positive basis momentum is steeper when the curve is already in backwardation (0.078 against 0.041; t-statistic of the difference of 2.5). It is also steeper when commercial hedgers are net short and thus pay for insurance (0.063 against 0.019; t-statistic of 2.1). Periods of a high CBOE Volatility Index (VIX), in which financial intermediaries have less capacity to bear risk, do not change the slope. Other implications of storage are

not borne out with statistical precision. The asymmetry does not differ significantly between storable commodities, precious metals, and livestock and dairy, and the location of the kink cannot be shown to shift with the state of the curve. We therefore treat scarcity as an interpretation that the evidence supports, not as an identified mechanism.

The one-sided information has economic value. Estimated in real time over 1990–2024, a position with separate slopes for positive and negative signals earns an alpha of 0.30% per month (t-statistic of 3.0) on linear basis momentum and the long-only portfolio. The alpha is 0.25% (t-statistic of 2.4) when time-series momentum and cross-sectional commodity factors are added. Time-series strategies gain from a net long exposure (Goyal and Jegadeesh, 2018), and about two thirds of the raw gain of the asymmetric position over linear basis momentum comes from its average net long exposure. The alpha measures the rest. In raw terms, the Sharpe ratio rises from 0.63 to 0.86 (significant at the 10% level) and, after 2000, from 0.40 to 0.77 (at the 5% level). The position trades no more than linear basis momentum. A rule that holds positions only on positive signals, the rule implied by a zero slope on negative signals, earns a similar alpha (0.33% per month).

The asymmetry weakens in recent years. Over 2010–2024, the slope on positive basis momentum halves, and the difference between the two slopes is not significant (t-statistic of 0.8). The weakening begins before basis momentum first circulated as a working paper in 2015, so it is not a publication effect.

The paper contributes to three literatures. First, it changes the reading of an established futures signal. The literature on commodity term-structure signals models their relation to returns linearly or through sorts (Erb and Harvey, 2006; Szymanowska et al., 2014; Koijen et al., 2018; Boons and Porras Prado, 2019; Bakshi et al., 2019; Paschke, Prokopczuk, and Wese Simen, 2020; Bianchi, Fan, Miffre, and Zhang, 2023). We show that the information of basis momentum about the level of the curve is one-sided, whereas its information about the spread comes mainly from the negative side. In concurrent work, Gu et al. (2025) argue from

the same convexity of the convenience yield that positive, but not negative, relative basis predicts returns in the cross-section. Relative basis measures the level of the curve and basis momentum its change, and the two are only weakly correlated (0.22). In our time-series design, neither side of relative basis predicts returns at the 5% level, and controlling for it leaves the asymmetry of basis momentum intact.

Second, we contribute to the debate on the economic source of the basis-momentum premium. Boons and Porras Prado (2019) conclude that basis momentum is unlikely to reflect storage, because its average effect is similar across storable and less storable commodities and because it also predicts returns in financial futures. Our evidence qualifies this conclusion without overturning it. Price pressure from limited intermediary capacity affects positive and negative signals alike (Hazelkorn, Moskowitz, and Vasudevan, 2023) and is consistent with the spread information, which does not differ significantly between the two sides. The additional, one-sided information about the level of the curve depends on the state of the curve and on the positions of hedgers (Keynes, 1930; Hirshleifer, 1988; De Roon, Nijman, and Veld, 2000; Kang, Rouwenhorst, and Tang, 2020), as scarcity implies.

Third, we contribute to the evidence on time-series predictability in futures markets (Moskowitz, Ooi, and Pedersen, 2012; Huang, Li, Wang, and Zhou, 2020; Hurst, Ooi, and Pedersen, 2017). The profits of equity anomalies come largely from the short leg (Stambaugh, Yu, and Yuan, 2012). In basis momentum, the short side of a time-series position earns nothing, while long positions carry the profits, as in several anomalies across asset classes (Israel and Moskowitz, 2013). The shape also matters for machine-learning approaches to nonlinear momentum. A flexible estimator recovers it only when it estimates a conditional expectation. With the Sharpe-ratio loss or the symmetrized training data common in the literature on nonlinear momentum (Lim, Zohren, and Roberts, 2019; Moskowitz, Sabbatucci, Tamoni, and Uhl, 2025), a neural network returns a nearly symmetric shape.

Section 2 states the predictions, and Section 3 describes the data and the empirical design.

Section 4 documents the asymmetry, Section 5 examines which information each side carries and why, and Section 6 evaluates its value in real time. Section 7 concludes.

2 Why Basis Momentum Should Be Asymmetric

The futures basis equals the cost of carrying inventory, interest and storage, net of the convenience yield (Kaldor, 1939; Working, 1949; Brennan, 1958). The convenience yield is a decreasing and convex function of inventories (Fama and French, 1988; Gorton et al., 2013). When inventories are ample, an additional unit is worth little and the curve is in contango. Arbitrage then bounds the contango at full carry (Working, 1949). When inventories become scarce, the non-negativity constraint on inventories comes close to binding, the convenience yield rises steeply, and the curve moves into backwardation (Deaton and Laroque, 1992; Routledge et al., 2000). In models of competitive storage, stockouts produce price spikes and positive skewness (Deaton and Laroque, 1992, 1996). Empirically, low inventories coincide with backwardation, higher volatility, and higher risk premiums (Gorton et al., 2013; Symeonidis, Prokopczuk, Brooks, and Lazar, 2012).

Basis momentum turns movements of the curve into a signal. It is positive when the front of the curve has outperformed the next contract, that is, when the curve has moved toward backwardation, and negative when the curve has moved toward contango. The convexity of the convenience yield implies that these moves differ in what they reveal. A move toward backwardation signals that the market has entered, or is approaching, a regime of scarcity. In this regime, the risk premium that hedgers pay to the speculators who absorb their hedging demand is high (Keynes, 1930; Hirshleifer, 1990; De Roon et al., 2000; Acharya, Lochstoer, and Ramadorai, 2013), and the regime persists. A move toward contango signals abundance. The curve is then pinned near full carry, and further moves carry little information about the risk premium of the commodity.

The convexity yields testable predictions. The slope on positive basis momentum should exceed that on negative basis momentum, and the relation should change slope near the sign of the signal, where moves toward contango turn into moves toward backwardation. The positive-side slope should be steeper when the curve is already in backwardation and when commercial hedgers are net short, because hedgers then pay for insurance (Ekeland, Lautier, and Villeneuve, 2019). Scarcity raises the price of every contract that matures before new supply arrives, so positive basis momentum should predict the level of the front of the curve rather than only the spread between its first two contracts. In its narrow form, the theory of storage also implies that the asymmetry is strongest where inventories can be held. Scarcity can arise without storage, for example when production capacity binds (Kogan, Livdan, and Yaron, 2009) or when the lease market for precious metals tightens (Barone-Adesi, Geman, and Theal, 2009). We note this possibility but do not test it.

Three alternatives make different predictions. First, the long side of time-series strategies may dominate because commodity futures earn a positive average return (Huang et al., 2020). The asymmetry should then vanish once each market's mean return is removed, and it should appear in every time-series commodity signal. Relatedly, time-series strategies gain from a net long exposure (Goyal and Jegadeesh, 2018), so the economic value of a long-biased position must be measured beyond the commodity premium. Second, Boons and Porras Prado (2019) interpret basis momentum as compensation for price pressure and volatility risk when speculators and intermediaries cannot absorb hedging demand. Price pressure should affect positive and negative signals alike (Hazelkorn et al., 2023). It does not imply that the shape depends on the state of the curve or that it concerns the level rather than the spread. Third, portfolio theory implies that positions should flatten for extreme signals. This produces a symmetric, S-shaped response (Moskowitz et al., 2025), not a one-sided kink.

3 Data and Empirical Design

3.1 Data

The dataset consists of 45 US dollar-denominated commodity futures in six sectors: energy (8), grains and oilseeds (12), livestock and dairy (7), industrial metals (8), precious metals (4), and softs (6). For each market, we collect end-of-day settlement prices of the first-, second-, and third-nearby contracts from Bloomberg (Bloomberg L.P., 2024), both roll-adjusted and unadjusted, and the positions of commercial traders from the Commitments of Traders reports of the CFTC, ICE, and the LME (U.S. Commodity Futures Trading Commission, 2024; ICE Futures Europe, 2024; London Metal Exchange, 2024). For the state tests, we use the VIX and, before 1990, the VXO (Chicago Board Options Exchange, 2026), and the recession dates of the NBER (National Bureau of Economic Research, 2024). Exhibit A1 of the Online Appendix lists the contracts. We separate industrial from precious metals because precious metals are held in large above-ground stocks.

Returns are computed from Bloomberg's roll-adjusted generic first- and second-nearby series, which hold the nearest contract and roll into the next contract at expiry. The excess return of a fully collateralized position is

$$r_{i,t+1} = \frac{F_{i,t+1}(T) - F_{i,t}(T)}{F_{i,t}(T)}, \tag{1}$$

where $F_{i,t}(T)$ is the price at the end of month t of the contract on commodity i that expires at T, so that a roll never mixes two contracts in one return. We denote by $r_{i,t+1} \equiv r^{(1)}_{i,t+1}$ the excess return of the first-nearby and by $r^{(2)}_{i,t+1}$ that of the second-nearby contract, and by $F^{(n)}_{i,t}$ the unadjusted price of the n-th nearby contract. Unadjusted prices are used only for the state of the curve at a point in time, namely the basis $\ln F^{(1)}_{i,t} - \ln F^{(2)}_{i,t}$ and the relative basis. They are never used for returns.

We apply four data screens. First, for two markets, Bloomberg's adjusted series coincide with the unadjusted ones and therefore contain the roll gaps. For the old NYMEX unleaded gasoline contract (1986–2006), we compute returns from the unadjusted first- to third-nearby prices on the NYMEX roll calendar; the same procedure reproduces Bloomberg's adjusted returns of its successor contract. Lumber is excluded, because its roll dates cannot be recovered. Second, prices that remain unchanged for 20 or more consecutive business days are treated as missing. Third, gaps are never filled with later prices. Fourth, trader positions enter with the last report published at least seven days before the end of the month in which a position is formed, and a report older than four weeks is not used. All signals are therefore known at the end of month t, before the return of month $t+1$ is earned.

Prices start in August 1959. The number of markets grows from 6 in 1960 to 30 in 1990 and 41 in 2010, and all six sectors are present from 1989 (Exhibit A2 and Exhibit A3 of the Online Appendix). Because the signal requires almost two years of returns, the first predictive regression month is July 1961. We refer to the regression sample as 1960–2024.

3.2 Basis Momentum and the Time-Series Signal

Following Boons and Porras Prado (2019), basis momentum is the difference between the 12-month cumulative returns of the first- and the second-nearby contract:

$$BM_{i,t} = \prod_{k=t-11}^{t} \left(1 + r_{i,k}\right) - \prod_{k=t-11}^{t} \left(1 + r_{i,k}^{(2)}\right). \tag{2}$$

It is available for 43 of the 45 markets, because propane and pork bellies lack a second-nearby series. Basis momentum differs from price momentum, which measures the level of past returns, and from the basis, which measures the current slope of the curve rather than its change.

Boons and Porras Prado (2019) implement basis momentum cross-sectionally. Because a cross-sectional rank removes the sign of the signal, our question requires its time-series version. We standardize each market's signal by its own volatility and express each market's return per unit of its ex-ante volatility:

$$s_{i,t} = \frac{BM_{i,t}}{\hat{\sigma}_{12}(BM)_{i,t}}, \qquad RaR_{i,t+1} = \frac{r_{i,t+1}}{\hat{\sigma}_{12}(r)_{i,t}}, \tag{3}$$

where $\hat{\sigma}_{12}$ is the standard deviation over the trailing 12 months, so that both quantities are known at the end of month t. We call $s_{i,t} > 0$ positive and $s_{i,t} < 0$ negative basis momentum. A strategy holds a position $w_{i,t} = f(s_{i,t})$ in each market, rebalances monthly, and earns

$$r^{p}_{t+1} = \frac{1}{N_t} \sum_{i=1}^{N_t} f(s_{i,t})\, \sigma^{target}\, RaR_{i,t+1}, \tag{4}$$

where $\sigma^{target} = 40\%$ per year as in Moskowitz et al. (2012) and N_t is the number of markets with a signal. Linear basis momentum uses $f(s) = s$. Nonlinear positions are rescaled so that their root mean square in the estimation sample equals that of s; the level of the positions does not affect Sharpe ratios.

3.3 Predictive Regressions and Inference

The asymmetric regression splits the signal at zero:

$$RaR_{i,t+1} = a_i + b^{+} s^{+}_{i,t} + b^{-} s^{-}_{i,t} + \varepsilon_{i,t+1}, \qquad s^{+} = \max(s, 0),\ s^{-} = \min(s, 0). \tag{5}$$

The linear benchmark restricts $b^{+} = b^{-} = b$:

$$RaR_{i,t+1} = a_i + b\, s_{i,t} + \varepsilon_{i,t+1}. \tag{6}$$

The market fixed effects a_i absorb each market's average return, so that the slopes measure predictability net of the commodity premium. A Wald test of $b^+ = b^-$ tests for an asymmetry. Standard errors are clustered by month to account for the cross-sectional correlation of commodity returns (Petersen, 2009). Bootstrap tests resample months in stationary blocks with a mean length of six months and keep all markets of a month together (Politis and Romano, 1994). Where a null hypothesis is imposed, we use a wild bootstrap that multiplies the residuals of all markets in a month by the same standard normal draw. Time-series alphas use Newey and West (1987) standard errors with six lags, and regressions over overlapping horizons use Driscoll and Kraay (1998) standard errors. Because many signals have been tested on commodity returns, we hold the central tests to the hurdle of a t-statistic of three (Harvey, Liu, and Zhu, 2016).

3.4 Locating the Change in Slope

Splitting at zero imposes the location of the change in slope. We therefore also estimate the location with a regression kink with an unknown threshold (Hansen, 2017):

$$RaR_{i,t+1} = a_i + \beta\, s_{i,t} + \delta\,(s_{i,t} - \gamma)_+ + \varepsilon_{i,t+1}. \tag{7}$$

The relation is continuous, with slope β below the kink γ and slope $\beta + \delta$ above it. We estimate γ by minimizing the sum of squared residuals over a grid between the 5th and 95th percentiles of the signal and report a 90% interval from the month bootstrap. The standard errors of the slopes account for the estimation of γ (Hansen, 2017). We test the null of no kink, $\delta = 0$, with the supremum of the F-statistic over the grid. Because γ is not identified under the null, the critical values come from a fixed-regressor wild bootstrap by month (Hansen, 1996).

3.5 Use of Generative AI

We used Claude (Anthropic), a generative AI tool based on a large language model, for two tasks: to assist with programming the Python code that constructs the data, runs the analyses, and produces the tables and figures, and to edit the text. The tool was not used for any other purpose. We reviewed all code and text and take full responsibility for the content of the article.

4 The Hockey Stick in Basis-Momentum Predictability

4.1 Model-Free Evidence

Is the relation between basis momentum and next-month returns linear? Exhibit 1 sorts all market-months into quantile bins of the signal and plots the mean risk-adjusted return of the following month, net of each market's mean return.

It is not. Across the bins of negative signals, expected returns show no trend, and none of these bins lies significantly above the market mean. Across the bins of positive signals, expected returns rise. The rise is concentrated in large moves toward backwardation: the three highest of the twenty bins, with signals above about two standard deviations, lie significantly above the market mean. Exhibit A5 of the Online Appendix shows the same pattern before and after 2000. In both halves, the highest decile of the signal is followed by returns significantly above the market mean, whereas the deciles of negative signals lie close to or slightly below it, without the steady decline that a linear relation implies. Because returns are measured net of each market's mean, the flat branch lies slightly below zero when the positive tail earns more than the mean.

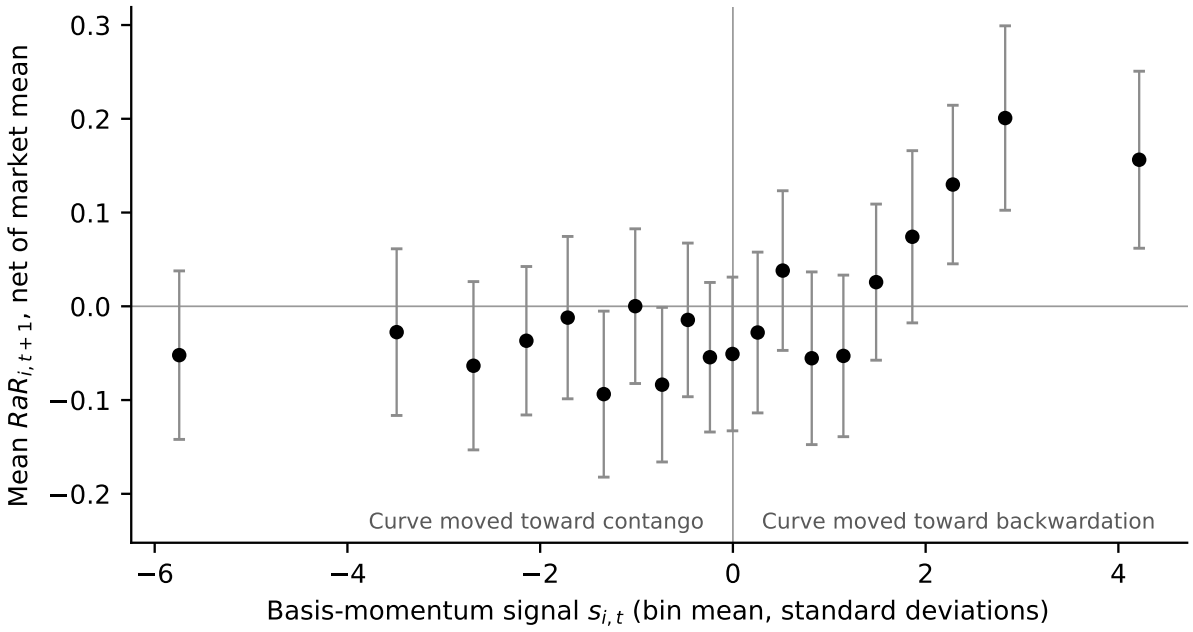


Exhibit 1: **Returns Rise after Moves toward Backwardation, Not after Moves toward Contango.** After negative basis momentum, the next month's risk-adjusted return does not vary with the signal. After positive basis momentum, it rises, and the three highest of the twenty bins, with signals above about two standard deviations, lie significantly above the market mean. The figure shows the mean of $RaR_{i,t+1}$ net of each market's mean in 20 pooled quantile bins of the signal $s_{i,t}$, 1960–2024, with 95% confidence intervals (standard errors clustered by month).

4.2 Positive and Negative Basis Momentum

Exhibit 2 turns the picture into tests.

The slope on positive basis momentum is 0.059 (t-statistic of 6.4), and the slope on negative basis momentum is 0.002 (t-statistic of 0.2). A one-standard-deviation increase in a positive signal raises the next month's return by 5.9% of its monthly volatility; a decrease in a negative signal has no effect. The difference between the slopes is significant over the full sample (t-statistic of 4.2), before 2000 (3.6), and after 2000 (2.7). It remains significant with standard errors clustered by month and market (t-statistic of 4.7) and with Driscoll and Kraay (1998) standard errors (3.8). With month fixed effects, which identify the slopes from the cross-section of each month, the asymmetry persists over the full sample (Wald p-value of 0.002). After 2000, the linear slope is insignificant, but the slope on positive basis

Exhibit 2: **The Shape of the Basis-Momentum Relation.**

	1960–2024	1960–1999	2000–2024
Linear: $RaR_{i,t+1} = a_i + b\, s_{i,t} + \varepsilon_{i,t+1}$			
b	0.026***	0.040***	0.011
	(5.08)	(5.40)	(1.63)
Asymmetric: $RaR_{i,t+1} = a_i + b^+ s^+_{i,t} + b^- s^-_{i,t} + \varepsilon_{i,t+1}$			
b^+	0.059***	0.087***	0.040***
	(6.40)	(5.65)	(3.41)
b^-	0.002	0.011	−0.011
	(0.24)	(0.98)	(−0.95)
$b^+ - b^-$	0.057***	0.076***	0.051***
	(4.17)	(3.55)	(2.73)
Wald $b^+ = b^-$, p-value	0.000	0.000	0.006
$t(b^+ - b^-)$, clustered by month and market	4.71	3.74	3.09
$t(b^+ - b^-)$, Driscoll–Kraay	3.80	3.30	2.73
Asymmetric with month fixed effects			
b^+	0.047***	0.073***	0.029***
b^-	0.009	0.013	0.001
Wald $b^+ = b^-$, p-value	0.002	0.002	0.091
Curvature: $RaR_{i,t+1} = a_i + b\, s_{i,t} + c\, s^3_{i,t} + \varepsilon_{i,t+1}$			
$c \times 1{,}000$	−0.10	−0.73***	0.12
	(−0.85)	(−3.34)	(1.00)
Markets	43	36	43
Observations	19,336	7,867	11,469

Pooled predictive regressions of the risk-adjusted return $RaR_{i,t+1} = r_{i,t+1}/\hat{\sigma}_{12}(r)_{i,t}$ on the basis-momentum signal $s_{i,t} = BM_{i,t}/\hat{\sigma}_{12}(BM)_{i,t}$, with market fixed effects a_i (and month fixed effects where stated). t-statistics in parentheses use standard errors clustered by month; the two rows below the Wald test report the t-statistic of $b^+ - b^-$ with standard errors clustered by month and market and with Driscoll and Kraay (1998) standard errors (12 lags); ***, **, * denote two-sided significance at the 1%, 5%, and 10% levels. The location of the kink is estimated in Exhibit 3.

momentum remains significant (t-statistic of 3.4): linear basis momentum lost its predictive power because its negative branch carries none. Under a null that imposes a linear relation, the 99% bootstrap critical value of the t-statistic of the difference is 2.22, far below the observed 4.17 (Exhibit A8 of the Online Appendix). Positive and negative basis momentum are therefore different predictors.

4.3 Where Does the Slope Change?

The asymmetric regression imposes a change in slope at zero. Exhibit 3 estimates its location with the regression kink of Equation (7).

Exhibit 3: **Where Is the Kink?**

	$\hat{\gamma}$	90% interval	Slope below	Slope above	p (no kink)	BM at kink (%)	Share above (%)
Panel A: Full sample and subperiods							
1960–2024	−0.10	[−2.90, 0.95]	0.001	0.058***	<0.001	−0.2	49
1960–1999	−2.90	[−4.11, 0.80]	−0.039	0.063***	0.001	−7.2	88
2000–2024	0.20	[−1.58, 1.50]	−0.010	0.043***	0.013	0.4	46
Panel B: Expanding estimation windows							
1960–1989	−3.00	[−3.75, 3.16]	−0.030	0.060***	0.054	−7.8	86
1960–1997	−2.85	[−4.21, 0.40]	−0.038	0.065***	0.001	−7.1	88
1960–2005	−0.35	[−2.80, 0.90]	0.000	0.074***	0.002	−0.9	50
1960–2013	−0.20	[−2.71, 0.20]	−0.004	0.068***	<0.001	−0.5	49
1960–2021	−0.15	[−2.81, 0.90]	−0.003	0.058***	<0.001	−0.3	50
Panel C: Storability							
Storable	−0.10	[−2.75, 0.95]	0.003	0.062***	0.001	−0.2	50
Precious metals	−0.30	[−2.95, 2.80]	−0.029	0.061**	0.125	−0.1	47
Livestock and dairy	2.60	[−3.70, 3.10]	0.009	0.096	0.425	10.0	9

Regression kink with an unknown threshold (Hansen, 2017), $RaR_{i,t+1} = a_i + \beta s_{i,t} + \delta(s_{i,t} - \gamma)_+ + \varepsilon_{i,t+1}$, with market fixed effects. $\hat{\gamma}$ minimizes the sum of squared residuals over the 5th to 95th percentiles of s; the interval is the 5th–95th percentile of $\hat{\gamma}$ over 199 stationary-bootstrap samples of months. Storable: energy, grains and oilseeds, industrial metals, and softs. Slopes below (β) and above ($\beta + \delta$) the kink; stars from t-statistics clustered by month that account for the estimation of γ (Hansen, 2017). p: test of no kink ($\delta = 0$) by the sup-F statistic with a fixed-regressor wild bootstrap by month (Hansen, 1996), 999 draws; <0.001: none of the draws exceeds the statistic. BM at kink: median of $\hat{\gamma}\,\hat{\sigma}_{12}(BM)_{i,t}$ in %, the 12-month return difference between the first- and second-nearby contract at which the slope changes. Share above: percentage of observations with $s_{i,t} > \hat{\gamma}$. Median skewness of BM across markets: storable 0.87, precious metals −1.07, livestock and dairy −0.18. ***, **, * denote significance at the 1%, 5%, and 10% levels.

The regression kink rejects a linear relation: none of 999 bootstrap draws exceeds the observed sup-F statistic. Over 1960–2024, the estimated location is −0.10 standard deviations, which corresponds to a difference of −0.2% between the 12-month returns of the first- and the second-nearby contract. The slope is 0.001 below the kink and 0.058 above it (t-statistic of 5.5, accounting for the estimation of the kink), and half of all observations lie above it. Above the kink, the relation is linear; adding the square of $(s_{i,t} - \hat{\gamma})_+$ does not improve the fit (t-statistic of −1.2).

The location, however, is estimated imprecisely. Its 90% interval runs from −2.90 to 0.95. Before 2000, and in the estimation windows through 1989 and 1997, the kink lies near −3 standard deviations, with more than 85% of the observations above it. In these samples, the relation is close to linear except for a flattening at the most negative signals, which the cubic term of Exhibit 2 also picks up. The kink near zero emerges once data after 2000 are included: from 2000 on, it lies at 0.20 ($p = 0.013$), and in the windows through 2005, 2013,

and 2021, between −0.35 and −0.15. The robust finding is that the two sides of the signal differ; that the slope changes exactly at zero holds for the full sample and the period after 2000, but not before.

4.4 Stability across Time, Commodities, and Specifications

We begin with stability over time, using the strategy that investors trade. Exhibit 4 splits the return of linear time-series basis momentum into the contributions of the positions taken on positive and on negative signals.

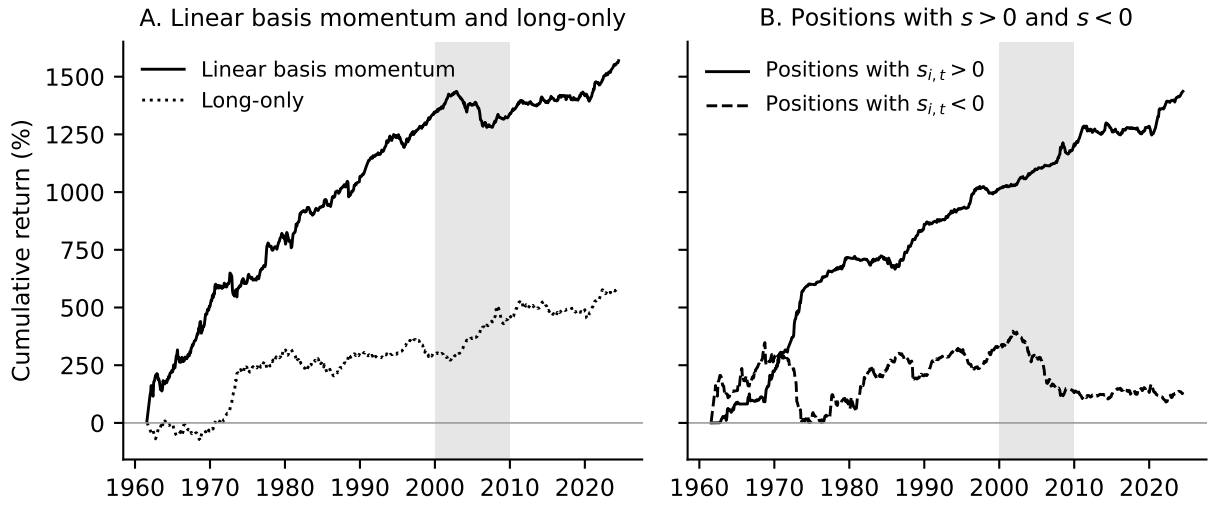


Exhibit 4: **Where Linear Basis Momentum Earns and Loses.** Panel A: cumulative return of the linear time-series basis-momentum strategy and of the long-only portfolio on the same markets, 1960–2024. Panel B: the linear strategy split into the contributions of the positions taken on positive and on negative signals; the two lines add up to the strategy in Panel A. Cumulative sums of monthly returns; shaded: 2000–2009.

The positions on positive signals earn a Sharpe ratio of 0.93 over 1960–2024 and are significantly profitable in every subperiod, including 2000–2009 (1.05). The positions on negative signals make up 55% of all positions. They earn a Sharpe ratio of 0.06 over the full sample and of −0.73 in 2000–2009; only in the 1990s did they pay (0.60). Because the linear strategy is short more often than long, it has a beta of −0.39 on the long-only portfolio and failed in the decade of the commodity boom (Exhibit A4 of the Online Appendix). Market

fixed effects remove this exposure from the predictive regression, yet the linear slope still vanishes in the 2000s. The failure is therefore not about the commodity premium but about the information in negative basis momentum.

Exhibit 5 varies the definition of the signal, the sample, and the universe one element at a time. Each variant addresses one concern.

Exhibit 5: **Robustness: Signal Definition, Sample, and Universe.**

	b^+	b^-	$t(b^+ - b^-)$	SR linear	SR asym.	$t(\alpha)$
Baseline	0.059***	0.002	4.17	0.63	0.86*	3.02
Panel A: Signal definition						
BM window 6 months	0.070***	0.012	2.88	0.75	0.88*	2.02
BM window 24 months	0.024***	0.004	2.24	0.43	0.63*	1.07
Volatility window 36 months	0.093***	−0.023*	5.33	0.58	0.78*	2.77
Expanding volatility of BM	0.110***	−0.021	4.54	0.68	0.85	2.53
Unscaled BM	0.598***	−0.080	3.13	0.60	0.72**	1.87
Signal clipped at 1st/99th percentiles	0.065***	0.000	4.40	0.63	0.85*	2.67
Panel B: Sample						
Without Energy	0.058***	0.002	3.86	0.48	0.78**	2.66
Without Precious Metals	0.059***	0.004	3.83	0.62	0.82	2.52
Without dairy, Ethanol, Iron Ore	0.061***	0.003	4.17	0.62	0.86*	2.98
From 1989	0.053***	−0.004	3.67	0.49	0.75	2.34
Without 2008-2009 and 2020	0.056***	0.005	3.64	0.62	0.89*	2.58
2010-2024	0.028**	0.010	0.82	0.75	0.77	1.18
2017-2024	0.035*	0.019	0.51	0.88	1.05	0.63
Panel C: Investable universe and weights						
S&P GSCI markets	0.060***	0.004	3.21	0.48	0.64	1.62
Constant universe from 1991	0.048***	0.001	2.45	0.54	0.76*	2.23
Sector-equal weights	0.059***	0.002	4.17	0.59	0.85**	2.77

Columns 2–4: asymmetric regression $RaR_{i,t+1} = a_i + b^+ s^+_{i,t} + b^- s^-_{i,t} + \varepsilon_{i,t+1}$, $s^+ = \max(s, 0)$, $s^- = \min(s, 0)$, with market fixed effects, standard errors clustered by month (stars: two-sided significance), and the t-statistic of $b^+ - b^-$. Columns 5–7: walk-forward strategies (February 1990–June 2024, or the stated sample): Sharpe ratios of linear basis momentum and of the asymmetric position (stars: one-sided bootstrap p-value of an improvement over linear basis momentum), and the Newey–West t-statistic of the asymmetric strategy's alpha on linear basis momentum and the long-only portfolio. The unscaled signal uses $BM_{i,t}$ without dividing by its volatility; with the 36-month window, both the signal and the returns are scaled by 36-month volatilities. From 1989: estimation and walk-forward training start in 1989, when all six sectors are present; 2010–2024 and 2017–2024: models trained on all earlier data, evaluated in the stated period. S&P GSCI markets: the 24 commodities of the index; constant universe: the markets with a signal in January 1991, 1991–2024; sector-equal weights: each sector receives the same weight in the strategies (the regression columns equal the baseline). ***, **, * denote significance at the 1%, 5%, and 10% levels.

The asymmetry does not depend on the 12-month window or on the volatility scaling of the signal: the difference between the slopes is significant with 6- or 24-month basis momentum, with 36-month or expanding volatility windows, with an unscaled signal, which addresses the concern that volatility scaling drives the profits of time-series strategies (Kim,

Tse, and Wald, 2016), and with a signal clipped at its 1st and 99th percentiles, which addresses extreme observations. It does not depend on a single sector or on crises: the difference is significant without energy, without precious metals, without the short histories of dairy, ethanol, and iron ore, from 1989, when all sectors are present, and without 2008–2009 and 2020. It holds in the 24 markets of the S&P GSCI, in a constant universe of the markets available since 1991, and with sector-equal weights. The slope on positive basis momentum is positive in all six sectors and significant in five, and that on negative basis momentum is insignificant in all six (Exhibit A7 of the Online Appendix). Market by market, the positive-side slope is positive in 33 of 42 markets and significant in 14; the negative-side slope is significant in 5, close to what chance implies. Within the 12-month window, the asymmetry stems from months $t-5$ to $t-1$ and $t-11$ to $t-6$ (t-statistics of the difference of 3.4 and 2.2), not from the most recent month (Exhibit A14).

The asymmetry weakens in recent years. Over 2010–2024, the slope on positive basis momentum is 0.028 (t-statistic of 2.0), half its full-sample value, and that on negative basis momentum is 0.010. Their difference is not significant (t-statistic of 0.8), and neither is it over 2017–2024 (0.5). The economic advantage weakens as well: since 2010, the asymmetric position earns a Sharpe ratio of 0.77 against 0.75 for linear basis momentum, and its alpha is insignificant. The weakening is not a publication effect in the sense of McLean and Pontiff (2016). It begins before the first working-paper version of Boons and Porras Prado (2019) in April 2015: from 2010 to April 2015, the difference between the slopes is -0.01, and the change after April 2015 relative to the earlier sample is not significant (t-statistic of -0.8; Exhibit A14). The timing is consistent with the inflow of financial investors into commodity futures, which lowers the futures premium (Baker, 2021), but the short recent sample cannot separate this explanation from a loss of power. The asymmetry is significant before 2000 (t-statistic of 3.6) and in 2000–2009 (2.9), and it cannot be established on 2010–2024 data alone.

4.5 Is the Asymmetry Specific to Basis Momentum?

If the hockey stick reflected a general feature of commodity futures, for example that short positions do not pay, every time-series commodity signal should show it. Exhibit 6 applies the same tests to the signals to which basis momentum is most closely related.

Exhibit 6: **Is the Asymmetry Specific to Basis Momentum?**

	Basis momentum	Momentum	Basis	Relative basis	Hedging pressure
Panel A: Asymmetric regression, sample of each signal					
c^+	0.059***	0.019***	0.042**	0.029	0.218*
	(6.40)	(3.93)	(2.17)	(1.54)	(1.95)
c^-	0.002	0.019**	0.028	−0.010	0.148
	(0.24)	(2.47)	(1.10)	(−0.43)	(0.94)
$t(c^+ - c^-)$	4.17	−0.03	0.41	1.21	0.33
Share $x > 0$	0.47	0.50	0.32	0.42	0.71
Start	1961-07	1961-07	1961-07	1961-07	1986-02
Observations	19,336	19,336	19,088	18,971	12,516
Panel B: Asymmetric regression, common sample 1986–2024					
c^+	0.054***	0.019***	0.011	0.020	0.200*
	(4.77)	(2.95)	(0.45)	(0.88)	(1.78)
c^-	−0.004	0.008	0.033	−0.050*	0.179
	(−0.40)	(0.82)	(0.89)	(−1.79)	(1.13)
$t(c^+ - c^-)$	3.46	0.83	−0.45	1.81	0.10
Panel C: Regression kink, sample of each signal					
$\hat{\gamma}$	−0.10	8.75	−1.50	−1.25	−0.20
Share above $\hat{\gamma}$	0.49	0.06	0.94	0.95	0.93
p-value, no kink	<0.002	0.553	0.004	0.184	0.914
Panel D: Sharpe ratios of the linear time-series strategy, 1960–2024					
All positions	0.71***	0.78***	0.31**	0.11	0.52***
Positions on $x > 0$	0.95***	0.71***	0.59***	0.37***	0.50***
Positions on $x < 0$	0.06	0.24**	−0.08	−0.22	0.02

Each column applies the tests of the paper to one signal $x_{i,t}$ on its own. Basis momentum: $s_{i,t}$. Momentum: the 12-1 month return divided by $\hat{\sigma}_{12}(r)_{i,t}$. Basis: $\ln F^{(1)}_{i,t} - \ln F^{(2)}_{i,t}$ and relative basis: $\ln F^{(1)}_{i,t} - 2\ln F^{(2)}_{i,t} + \ln F^{(3)}_{i,t}$ (Gu et al., 2025), both from unadjusted prices and divided by their expanding standard deviation. Hedging pressure: (short−long)/(long+short) of commercial traders. Panels A and B: $RaR_{i,t+1} = a_i + c^+ x^+_{i,t} + c^- x^-_{i,t} + \varepsilon_{i,t+1}$ with market fixed effects; t-statistics clustered by month in parentheses. Panel B uses the months and markets in which all five signals are available. Panel C: regression kink of Equation (7) with x in place of s; sup-F test with a fixed-regressor wild bootstrap by month (499 draws). Panel D: the linear strategy $w_{i,t} = x_{i,t}$ split into the contributions of the positions taken on positive and on negative signals (as in Panel D of Exhibit A4 of the Online Appendix); annualized Sharpe ratios, stars from one-sided stationary-bootstrap p-values of H_0: Sharpe ≤ 0. ***, **, * denote significance at the 1%, 5%, and 10% levels.

Only basis momentum is asymmetric. For time-series momentum, the slopes on positive and negative signals are identical (0.019 each). For the basis, the relative basis, and hedging pressure, the difference is insignificant, with t-statistics between 0.3 and 1.2 (at most 1.8 in the common sample). The regression kink rejects a linear relation for the basis, but at

−1.50 standard deviations, deep in contango and with 94% of the observations above it; it reflects the bound of full carry, not a change in slope at the sign of the signal. At the strategy level, the short positions of the curve signals also earn little (Panel D), because the positive average return of commodity futures favors long positions. Net of each market's mean return, however, only basis momentum splits into a predictive and a non-predictive branch.

Nor is the asymmetry a proxy for these signals. Basis momentum has a pooled correlation of 0.21 with time-series momentum, 0.32 with the basis, and close to zero with hedging pressure. With the positive and negative parts of all three as controls, the slope on positive basis momentum is 0.048 (t-statistic of 4.2), that on negative basis momentum remains zero, and the asymmetry remains significant (Exhibit A9 of the Online Appendix). Controlling for the positive and negative parts of the basis and the relative basis leaves the slopes on basis momentum unchanged (Panel C of Exhibit 8). The asymmetry is a property of the change in the curve, which basis momentum measures, rather than of its level, past returns, or positioning.

5 Which Information Does Each Side Carry?

The asymmetry concerns the direction of returns. This section asks which component of the futures return each side of the signal predicts, under which states of the market the positive side is most informative, and which predictions of the scarcity interpretation fail.

5.1 The Level and the Spread of the Futures Curve

The return of the nearby contract is the return of the second-nearby contract plus the spreading return, long the first- and short the second-nearby contract. Szymanowska et al. (2014) associate nearby returns mainly with the spot premium and spreading returns with

term premiums. Boons and Porras Prado (2019) show that basis momentum predicts both, and that the spreading premium comes mostly from the short leg. Exhibit 7 estimates the asymmetric regression for each component.

Exhibit 7: **The Level and the Spread of the Futures Curve: Which Return Carries the Asymmetry?**

	1960–2024			2000–2024			Kink	
Dependent return	b^+	b^-	$t(\Delta)$	b^+	b^-	$t(\Delta)$	$\hat{\gamma}$	p
Nearby, $r^{(1)}$	0.059***	0.002	4.17	0.040***	−0.011	2.73	−0.10	<0.005
	(6.40)	(0.24)		(3.41)	(−0.95)		[−2.90, 1.00]	
Second nearby, $r^{(2)}$	0.061***	−0.005	4.81	0.048***	−0.015	3.45	−0.10	<0.005
	(6.63)	(−0.65)		(4.12)	(−1.36)		[−1.46, 0.90]	
Spreading, $r^{(1)} - r^{(2)}$	0.020*	0.033***	−0.80	−0.004	0.029**	−1.49	2.60	0.251
	(1.91)	(3.84)		(−0.31)	(2.29)		[−2.67, 3.10]	

Asymmetric regressions $y_{i,t+1} = a_i + b^+ s^+_{i,t} + b^- s^-_{i,t} + \varepsilon_{i,t+1}$, where y is the return of the first-nearby contract (the baseline RaR), of the second-nearby contract, or the spreading return (long the first-, short the second-nearby contract), each divided by its own trailing 12-month volatility at t. Before scaling, the nearby return is the sum of the second-nearby and the spreading return; Szymanowska et al. (2014) interpret nearby returns as dominated by the spot premium and spreading returns as term premiums. Market fixed effects; t-statistics clustered by month in parentheses; $t(\Delta)$: Wald test of $b^+ = b^-$. Kink: regression kink with unknown threshold as in Exhibit 3 (1960–2024), 90% bootstrap interval in brackets, p: sup-F test of no kink, 199 draws. ***, **, * denote significance at the 1%, 5%, and 10% levels.

The two components behave differently. Positive basis momentum predicts the second-nearby return as strongly as the nearby return (slopes of 0.061 and 0.059, t-statistics of 6.6 and 6.4), and negative basis momentum predicts neither. For the second-nearby return, the difference between the slopes is larger than for the nearby return (t-statistic of 4.8), and the kink again lies near zero. The spreading return is different. It is predicted mainly by negative basis momentum (0.033, t-statistic of 3.8); positive basis momentum predicts it only weakly (0.020, t-statistic of 1.9), and the difference between the two slopes is not significant (t-statistic of −0.8). After 2000, the spreading return is predicted by the negative side (0.029, t-statistic of 2.3), again without a significant difference between the two slopes. The second-nearby return never contains the last days before a contract's expiry, so the asymmetry does not arise from expiry effects in the nearby contract.

Basis momentum therefore combines two pieces of information. Its information about the spread between the first two contracts comes mainly from the negative side and does not differ significantly between the two sides. Its information about the level of the front of the

curve is one-sided: a move toward backwardation forecasts a higher level, consistent with a spot premium that rises with scarcity, whereas a move toward contango forecasts only that the nearby contract will underperform the next one. A time-series position in the nearby contract trades the level, and its negative side therefore earns nothing.

The cross-section confirms that negative basis momentum carries relative information. In the cross-sectional factor of Boons and Porras Prado (2019), markets with the lowest basis momentum underperform the average market (alpha of 0.40% per month, t-statistic of 4.2, 1960–2024), and markets with the highest basis momentum outperform it (0.56%, t-statistic of 6.2; Exhibit A10 of the Online Appendix). Markets whose curve has moved toward contango do worse than other markets, but they do not fall in absolute terms. The decomposition reconciles our results with the evidence on strategies that trade the shape of the curve (Paschke et al., 2020; Bianchi et al., 2023): the information that negative basis momentum carries belongs to spread and relative positions, not to directional shorts.

5.2 The State of the Curve and the Positions of Hedgers

If scarcity makes the positive side informative, the positive-side slope should be steepest where scarcity is most likely. We measure the state of the curve by the sign of $\ln F_{i,t}^{(1)} - \ln F_{i,t}^{(2)}$ from unadjusted prices; backwardation occurs in 32% of all market-months. We measure hedging pressure by $HP = (\text{short} - \text{long})/(\text{long} + \text{short})$ of commercial traders. Exhibit 8 reports the results.

Both states sharpen the positive side. When the curve is already in backwardation, the slope on positive basis momentum is 0.078 (t-statistic of 6.8); in contango, it is 0.041 (t-statistic of 3.4), and the difference is significant (t-statistic of 2.5). Commercial hedgers are net short in 71% of the observations. In these observations, the slope is 0.063 (t-statistic of 4.9); when hedgers are net long, it is 0.019 and insignificant, and the difference is significant (t-statistic of 2.1). In neither state does negative basis momentum predict lower returns.

Exhibit 8: **Basis Momentum, the State of the Curve, and Hedging Pressure.**

	Panel A: Curve		*Panel B: Hedgers*	
	Backwardation	Contango	Net short	Net long
b^+	0.078***	0.041***	0.063***	0.019
	(6.77)	(3.36)	(4.91)	(1.03)
b^-	−0.033*	0.009	−0.003	−0.010
	(−1.88)	(1.05)	(−0.25)	(−0.70)
b^+ in state minus out of state	0.036** (2.45)		0.043** (2.05)	
Share of observations in state	0.32		0.71	
Kink $\hat{\gamma}$	−0.75	1.05	−1.30	1.35
p (no kink)	0.015	0.035	0.010	0.266

Panel C: Horse race with the level of the curve			
	BM only	+ Relative basis	+ Relative basis + basis
s^+	0.061*** (6.59)	0.061*** (6.44)	0.058*** (6.10)
s^-	0.003 (0.35)	0.003 (0.42)	0.002 (0.29)
Relative basis$^+$		0.004 (0.20)	−0.022 (−0.76)
Relative basis$^-$		−0.015 (−0.70)	−0.020 (−0.85)
Basis$^+$			0.030 (1.04)
Basis$^-$			0.017 (0.65)
$t(b^+ - b^-)$ for basis momentum	4.20	4.13	3.97

Panels A and B: asymmetric regression $RaR_{i,t+1} = a_i + b^+ s^+_{i,t} + b^- s^-_{i,t} + \varepsilon_{i,t+1}$ with slopes interacted with the state at t. Panel A: backwardation, $\ln F^{(1)}_{i,t} - \ln F^{(2)}_{i,t} > 0$, from unadjusted first- and second-nearby prices. Panel B: hedging pressure of commercial traders, $HP = (\text{short} - \text{long})/(\text{long} + \text{short})$, $HP > 0$ (hedgers net short), from the last report at least seven days before the end of month t (CFTC, ICE, LME; from 1986). The kink rows estimate the regression kink of Exhibit 3 within each state (p: sup-F test with wild bootstrap, 199 draws). Panel C: horse race with the positive and negative parts of the relative basis $(\ln F^{(1)} - \ln F^{(2)}) - (\ln F^{(2)} - \ln F^{(3)})$ (Gu et al., 2025) and of the basis, each divided by its expanding standard deviation per market (common sample). Market fixed effects; t-statistics clustered by month in parentheses; ***, **, * denote significance at the 1%, 5%, and 10% levels.

When hedgers are net short, a move of the curve toward backwardation signals both scarcity and a high premium for the insurance that hedgers buy (Hirshleifer, 1988; De Roon et al., 2000; Ekeland et al., 2019). The positive-side slope persists for at least a year (Section 5.3), which matches the persistent hedging premium rather than the short-lived premium for accommodating changes in hedgers' positions (Kang et al., 2020).

The risk-bearing capacity of financial intermediaries does not account for the state dependence of the positive side. In periods of high VIX, financial traders reduce their positions and push risk back to hedgers (Cheng, Kirilenko, and Xiong, 2015). Interacting the slopes with a high-VIX state (from 1989) leaves the positive-side slope unchanged (t-statistic of the interaction of 1.2), whereas the backwardation interaction remains significant when both are included (t-statistic of 2.1; Exhibit A14 of the Online Appendix).

Three refinements of the state tests are not significant. Splitting each market at its own median hedging pressure gives the same ordering (0.057 against 0.034), but the difference is not significant (t-statistic of 1.2). Interacted with the standardized basis as a continuous variable, the positive-side slope does not change (t-statistic of -0.4); with hedging pressure as a continuous variable, the interaction is positive and marginally significant (t-statistic of 1.8; Exhibit A11). The point estimates of the kink order as scarcity implies, at -0.75 in backwardation and at $+1.05$ in contango, but a month-block bootstrap of this shift gives a 90% interval from -2.65 to 4.15 ($p = 0.15$), and the shift between hedgers net long and net short is not significant either ($p = 0.30$).

5.3 Commodity Groups, Skewness, and Persistence

In its narrow form, the theory of storage implies that the asymmetry is strongest where inventories can be held. Panel C of Exhibit 3 estimates the regression kink separately for storable commodities (energy, grains and oilseeds, industrial metals, and softs), precious metals, and livestock and dairy. In storable commodities, the slope is zero below the kink and 0.062 above it, and the test rejects a linear relation ($p = 0.001$). For precious metals, the slope above the kink is similar, but the kink is not significant ($p = 0.13$); gold and silver are in backwardation in only 1% and 0.5% of the months. For livestock and dairy, the test finds no kink ($p = 0.43$). The point estimates thus order the groups as storage predicts, but the data do not distinguish them. In one regression with group-specific slopes, the asymmetry in storable commodities does not differ significantly from that in precious metals (t-statistic of -0.6) or in livestock and dairy (0.6), and the two other groups taken together show a significant asymmetry of their own (t-statistic of 2.0; Exhibit A11 of the Online Appendix). With four and six markets, these groups are small. This implication is not supported, and storability does not discriminate between the interpretations.

The distribution of the signal is consistent with stockouts. In models of competitive

storage, stockouts produce rare, large price spikes, whereas arbitrage bounds the curve at full carry (Deaton and Laroque, 1992, 1996). Pooled across markets, the 99th percentile of basis momentum is +40%, its 1st percentile −26%, and its skewness 1.22. The median skewness is 0.87 across storable commodities and positive for 77% of them; it is −1.07 for precious metals and −0.18 for livestock and dairy. Positive basis momentum predicts returns where large moves toward backwardation occur, which matches the concentration of the predictive content in the upper tail of the signal (Exhibit 1).

Scarcity also persists, because inventories are rebuilt only slowly. The predictive slope on positive basis momentum declines only gradually, from 0.059 per month at a one-month horizon to 0.035 per month at twelve months, with t-statistics between 4.5 and 7.0. Negative basis momentum predicts returns at no horizon, and the difference between the slopes is significant at every horizon (t-statistics of 3.2 or more; Exhibit A13 of the Online Appendix). The slope on positive basis momentum is similar in and out of NBER recessions and periods of high commodity volatility (Exhibit A12).

Taken together, the evidence supports a one-sided information content of basis momentum that is consistent with scarcity: the positive side predicts the level of the curve, its slope is steeper in backwardation and when hedgers are net short, and its predictive power persists. It does not identify scarcity against price pressure. Three implications of storage, a stronger asymmetry in storable commodities, a kink that shifts with the state of the curve, and a slope that rises continuously with the basis, are not borne out with statistical precision.

6 Economic Value and Implementability

6.1 Real-Time Design

Can an investor who estimates the shape with past data only exploit it? We compare three positions, each formed only with information available at the time. The first is linear

basis momentum, $f(s) = s$. The second is the asymmetric position $f(s) \propto \hat{b}^{+}s^{+} + \hat{b}^{-}s^{-}$, with $\hat{b}^{\pm}$ estimated by pooled least squares on market-demeaned returns. The third is the positive-only rule $f(s) = \max(s, 0)$, which requires no estimation and corresponds to a zero slope on negative signals. We re-estimate on expanding windows with data through December 1989, 1997, 2005, 2013, and 2021 and apply each estimate until the next re-estimation (walk-forward, February 1990–June 2024). Alternatively, we estimate once with data through December 1999 (single split, February 2000–June 2024). We evaluate the strategies by their Sharpe ratios, bootstrap tests of Sharpe-ratio differences, and alphas on linear basis momentum, the long-only portfolio ($f(s) = 1$ on the same markets), time-series momentum (the sign of the return from month $t - 12$ to $t - 1$; Moskowitz et al., 2012), and cross-sectional basis-momentum, momentum, carry, and hedging-pressure factors. For comparison, Exhibit 9 also reports the long-only portfolio, time-series momentum, and the neural-network position of Section 6.4.

6.2 Real-Time Value

Exhibit 9 reports the results.

The asymmetric position earns alpha beyond the commodity premium and linear basis momentum. Over 1990–2024, its alpha on linear basis momentum and the long-only portfolio is 0.30% per month (t-statistic of 3.0). It is 0.24% (t-statistic of 2.4) when time-series momentum is added and 0.25% (t-statistic of 2.4) when the cross-sectional factors are added as well. In the single split, which covers the period in which linear basis momentum is weakest, the alpha after all factors has a t-statistic of 2.5.

The raw Sharpe ratio overstates the gain. The asymmetric position is long more often than short and has a beta of 0.91 on the long-only portfolio. Time-series strategies gain from a net long exposure (Goyal and Jegadeesh, 2018). Splitting each strategy's return into a cross-sectional part, a static net long part, and a timing part shows that about two thirds

Exhibit 9: **Exploiting the Asymmetry in Real Time.**

	Linear BM	Asymmetric	Neural net	s^+ only	Long-only	TSMOM
Walk-forward, February 1990–June 2024						
Sharpe ratio	0.63***	0.86***	0.83***	0.85***	0.43**	0.67***
Δ Sharpe vs. linear BM		0.23*	0.20*	0.22	−0.20	0.04
α on linear BM, long-only		0.30***	0.34**			
		(3.02)	(2.17)			
α on linear BM, long-only, TSMOM		0.24**	0.23			
		(2.35)	(1.52)			
α on the above and XS factors		0.25**	0.19			
		(2.38)	(1.19)			
α on the above and asymmetric			0.01			
			(0.07)			
Single split, February 2000–June 2024						
Sharpe ratio	0.40*	0.77***	0.26	0.83***	0.57**	0.45**
Δ Sharpe vs. linear BM		0.37**	−0.14	0.42*	0.16	0.04
α on linear BM, long-only		0.31***	0.03			
		(2.81)	(0.29)			
α on linear BM, long-only, TSMOM		0.29**	0.04			
		(2.55)	(0.33)			
α on the above and XS factors		0.29**	0.03			
		(2.50)	(0.23)			
α on the above and asymmetric			−0.09			
			(−0.94)			

Out-of-sample performance of equally weighted time-series strategies with return $w_{i,t}\,\sigma^{target} RaR_{i,t+1}$ in each market (Equation (4)). Linear BM: $w_{i,t} = s_{i,t}$. Asymmetric: $w_{i,t} \propto \hat{b}^+ s^+_{i,t} + \hat{b}^- s^-_{i,t}$ with $\hat{b}^\pm$ estimated by pooled OLS on market-demeaned RaR using data through December 1989, 1997, 2005, 2013, and 2021 (single split: December 1999). Neural net: ensemble of ten networks of the conditional expectation of market-demeaned RaR given s (mean-squared-error loss, Section 6.4) estimated on the same windows. Positions are scaled to the root mean square of s in the training window. s^+ only: $w_{i,t} = \max(s_{i,t}, 0)$, a rule without estimation. Long-only: $w_{i,t} = 1$; TSMOM: $w_{i,t}$ = the sign of the 12-month return skipping the most recent month (Moskowitz et al., 2012); all on the same markets and months. XS factors: cross-sectional tercile long-short portfolios of excess returns sorted on basis momentum (Boons and Porras Prado, 2019), 12-1 month momentum, the basis $\ln F^{(1)} - \ln F^{(2)}$ from unadjusted prices (Koijen et al., 2018), and the hedging pressure of commercial traders (from 1986; Basu and Miffre, 2013). Sharpe ratios are annualized; stars refer to one-sided stationary-bootstrap p-values (months resampled, mean block length 6) of H_0: Sharpe ≤ 0 and, for Δ Sharpe, of H_0: no improvement over linear BM. α in % per month from OLS on the stated strategies, Newey–West t-statistics (6 lags) in parentheses; ***, **, * denote significance at the 1%, 5%, and 10% levels.

of the raw gain over linear basis momentum, 0.61 of 0.89 percentage points per month, comes from the average net long position (Exhibit A14 of the Online Appendix). The long-only portfolio spans this part. The rest comes from the cross-sectional positions and from the timing of the net exposure (t-statistics of 1.7 and 2.0), and this is what the alpha measures. In raw terms, the Sharpe ratio rises from 0.63 to 0.86 in the walk-forward period, a difference that is significant at the 10% level, and from 0.40 to 0.77 after 2000, significant at the 5% level. Tests with the heteroskedasticity- and autocorrelation-robust standard errors of Ledoit and Wolf (2008) give the same conclusions. The asymmetric position beats linear basis momentum in each of the six sectors (Exhibit A15).

The positive-only rule earns a Sharpe ratio of 0.85 in the walk-forward period and of 0.83 after 2000, and an alpha of 0.33% per month (t-statistic of 3.3). It is not a mined rule: it is the position that the estimated slopes imply when the slope on negative signals is zero, and it follows from the information content documented in Section 5. Its gain in raw performance over linear basis momentum, however, comes entirely from its net long position; its cross-sectional part is weaker than that of linear basis momentum, because it gives up the relative information of negative signals. The estimated asymmetric position keeps a small short side and retains part of this information.

6.3 Costs, Turnover, and Risk

Exhibit 10 reports the implementation.

Exhibit 10: **Implementation and the Contribution of Positive and Negative Signals.**

	Linear BM	Asymmetric	Neural net	Long-only	TSMOM
Panel A: Turnover, costs, and risk					
Sharpe ratio, gross	0.63	0.86	0.83	0.43	0.67
Sharpe ratio, 5 bp	0.59	0.84	0.80	0.43	0.64
Sharpe ratio, 10 bp	0.56	0.81	0.76	0.42	0.62
Turnover per month	1.34	1.35	1.42	0.15	0.57
Skewness	−0.26	0.74	0.45	−0.22	−0.10
Maximum drawdown (%)	−67.7	−23.7	−29.5	−49.0	−40.1
Worst month (%)	−11.2	−10.4	−9.6	−11.9	−12.8
Panel B: Contribution of the positions with positive and negative signals					
Sharpe ratio, $s_{i,t} > 0$	0.85	0.84	0.85		
Sharpe ratio, $s_{i,t} < 0$	−0.10	−0.09	−0.15		
Share of mean return, $s_{i,t} > 0$	1.15	1.04	1.12		
Share of mean return, $s_{i,t} < 0$	−0.15	−0.04	−0.12		

Walk-forward strategies of Exhibit 9, February 1990–June 2024. Turnover is the average absolute monthly change of the notional exposure $w_{i,t}\sigma^{target}/\hat{\sigma}_{12}(r)_{i,t}$ per market. Net Sharpe ratios deduct 5 or 10 basis points per unit of notional traded; the asymmetric strategy's net Sharpe ratio stays above that of linear basis momentum for costs of at least 200 basis points. Maximum drawdown and worst month refer to the uncompounded strategy return, with each strategy scaled ex post to an annualized volatility of 10%. Panel B splits each strategy return into the contributions of the positions with positive and negative signals, each divided by the number of markets.

The asymmetric position trades as much as linear basis momentum, with a monthly turnover of 1.35 against 1.34 units of notional exposure per market. At 5 and 10 basis points per unit traded, its Sharpe ratio is 0.84 and 0.81, against 0.59 and 0.56 for linear basis

momentum. Its returns are positively skewed (0.74 against −0.26). At the same volatility, its maximum drawdown is about a third of that of the linear strategy (24% against 68%), and its worst month is no worse. In all strategies, the positions taken on positive signals earn the return, and the positions taken on negative signals lose (Panel B). Shorting a futures contract costs no more than buying it, so the short side is implementable; it is simply not compensated.

The main risk of the asymmetric position is a commodity crash. At 10% volatility, it lost between 1.0% and 4.2% per month in the commodity crash of 2008–2009, the oil-price collapse of 2014–2016, and the first months of COVID-19, about 60% to 80% of the losses of the long-only portfolio, and time-series momentum gained in the last two episodes (Exhibit A17 of the Online Appendix).

For long-only investors, the asymmetry does not raise the Sharpe ratio. Overweighting markets with positive and underweighting markets with negative basis momentum raises the out-of-sample Sharpe ratio of an equal-weighted commodity portfolio from 0.43 to 0.76, but a tilt by linear basis momentum does as well, because the long-only constraint itself removes the short side. The asymmetry raises the value of basis momentum for investors who can short, not for the long-only tilt.

6.4 Does a Flexible Estimator Find the Same Shape?

The regressions test shapes that we specify. To let the data choose the shape, we estimate the conditional expectation $E[RaR_{i,t+1} - \bar{r}_i \mid s_{i,t}] = g(s_{i,t})$ with a feed-forward neural network that imposes no functional form (Gu, Kelly, and Xiu, 2020). The network has a single input, the signal; its target is the risk-adjusted return net of the market's mean in the training window, so that a network that simply learned to be long would not reduce the loss. We use a mean-squared-error loss with a free intercept and select among 32 configurations by the mean squared error on the last four years of each training window, averaging ten networks

with different random seeds. Exhibit A18 of the Online Appendix reports the specification.

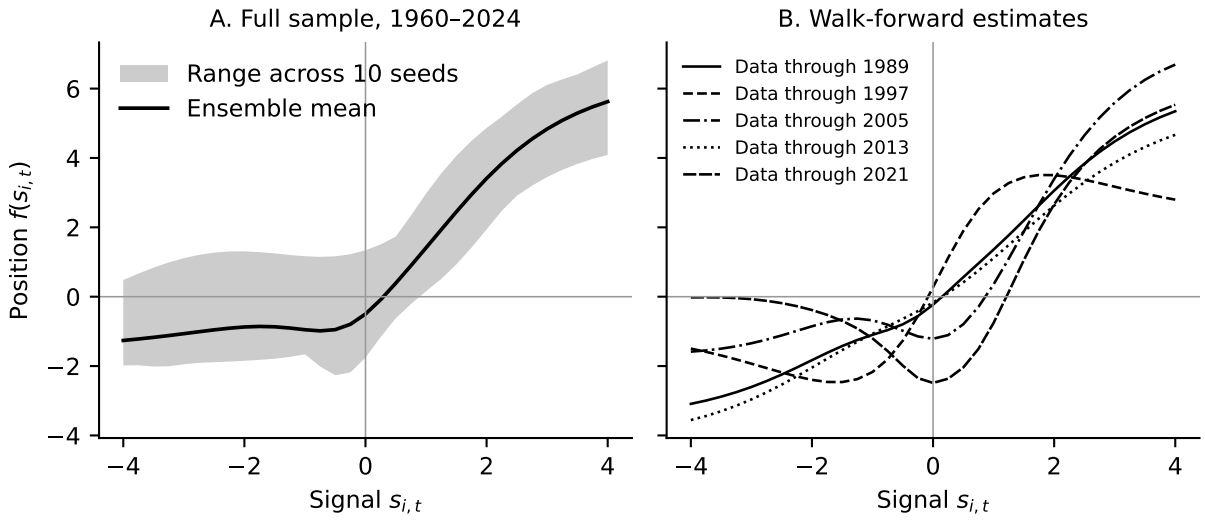


Exhibit 11: **The Shape Estimated by the Neural Network.** Network position $f(s_{i,t})$, that is, the estimated conditional expectation $\hat{g}(s_{i,t})$ rescaled to the root mean square of $s_{i,t}$ in the training window, plotted between the 1st and 99th percentiles of the signal. Panel A: full sample 1960–2024, mean and range across an ensemble of ten networks with different seeds. Panel B: ensemble means of the networks estimated on expanding windows with data through the stated year.

Estimated on the full sample, the network recovers the hockey stick (Exhibit 11): its response is nearly flat for negative signals and rises steeply for positive ones, with a slope for positive signals about eight times that for negative signals. In every walk-forward window, the response is steeper for positive than for negative signals, although in the windows through 1989, 1997, and 2013 it is closer to linear. The network finds the shape only when it estimates the conditional expectation. With a Sharpe-ratio loss, which optimizes a position rather than an expectation (Lim et al., 2019; Moskowitz et al., 2025), the median difference between the two slopes falls from 0.74 to 0.11. With symmetrized training data, which add each observation with the signs of the signal and the return reversed and thus impose $g(-s) = -g(s)$, it is -0.02 (Exhibit A21 of the Online Appendix). The network adds no value beyond the two-slope position: its walk-forward Sharpe ratio is 0.83, its alpha disappears once the asymmetric position is added as a control (t-statistic of 0.1), its out-of-sample R^2 is the lowest of all signal-based forecasts (Exhibit A20), and estimated once on data through

1999 it learns a nearly linear shape. The flexible estimator confirms the shape; the two-slope position extracts its value with less estimation noise.

7 Conclusion

Basis momentum is traded as a symmetric signal. Its information is not symmetric. A move of the futures curve toward backwardation forecasts a higher level of commodity futures prices; a move toward contango forecasts only that the nearby contract will underperform the next contract. Over 1960–2024, the short positions of time-series basis momentum earned a Sharpe ratio of 0.06, and they account for the failure of linear basis momentum in 2000–2009. Related signals do not share this asymmetry.

The positive side is most informative when the curve is already in backwardation and when commercial hedgers are net short, and its predictive power persists for a year. This is consistent with a scarcity premium that is bounded on the side of abundance. The evidence does not identify scarcity against price pressure: the asymmetry is not significantly stronger in storable commodities, and the location of the kink is imprecise. The asymmetry is weak and insignificant after 2010.

For futures investors, the result separates two uses of the same signal. Positive basis momentum belongs in directional long positions. Negative basis momentum belongs, if anywhere, in spread or cross-sectional positions, not in directional shorts. Estimated in real time, a position with separate slopes for the two sides earns an alpha of 0.30% per month beyond linear basis momentum and the commodity premium, at the same turnover. Flexible estimators recover this shape only if they estimate a conditional expectation rather than impose symmetry.

References

Acharya, V. V., L. A. Lochstoer, and T. Ramadorai (2013). Limits to arbitrage and hedging: Evidence from commodity markets. *Journal of Financial Economics 109*(2), 441–465.

Baker, S. D. (2021). The financialization of storable commodities. *Management Science 67*(1), 471–499.

Bakshi, G., X. Gao, and A. G. Rossi (2019). Understanding the sources of risk underlying the cross section of commodity returns. *Management Science 65*(2), 619–641.

Barone-Adesi, G., H. Geman, and J. Theal (2009). On the lease rate, the convenience yield and speculative effects in the gold futures market. Research Paper 09-07, Swiss Finance Institute.

Basu, D. and J. Miffre (2013). Capturing the risk premium of commodity futures: The role of hedging pressure. *Journal of Banking & Finance 37*(7), 2652–2664.

Bianchi, R. J., J. H. Fan, J. Miffre, and T. Zhang (2023). Exploiting the dynamics of commodity futures curves. *Journal of Banking & Finance 154*, 106965.

Bloomberg L.P. (2024). Commodity futures settlement prices: Generic first-, second-, and third-nearby contracts, roll-adjusted and unadjusted. Bloomberg Terminal. Monthly data, August 1959–June 2024; licensed, not redistributable.

Boons, M. and M. Porras Prado (2019). Basis-momentum. *The Journal of Finance 74*(1), 239–279.

Brennan, M. J. (1958). The supply of storage. *The American Economic Review 48*(1), 50–72.

Campbell, J. Y. and S. B. Thompson (2008). Predicting excess stock returns out of sample: Can anything beat the historical average? *The Review of Financial Studies 21*(4), 1509–1531.

Cheng, I.-H., A. Kirilenko, and W. Xiong (2015). Convective risk flows in commodity futures markets. *Review of Finance 19*(5), 1733–1781.

Chicago Board Options Exchange (2026). CBOE volatility index: VIX [VIXCLS] and CBOE S&P 100 volatility index: VXO [VXOCLS]. Retrieved from FRED, Federal Reserve Bank of St. Louis, https://fred.stlouisfed.org. Accessed 7 October 2026.

Clark, T. E. and K. D. West (2007). Approximately normal tests for equal predictive accuracy in nested models. *Journal of econometrics 138*(1), 291–311.

De Roon, F. A., T. E. Nijman, and C. Veld (2000). Hedging pressure effects in futures markets. *The Journal of Finance 55*(3), 1437–1456.

Deaton, A. and G. Laroque (1992). On the behaviour of commodity prices. *The Review of Economic Studies 59*(1), 1–23.

Deaton, A. and G. Laroque (1996). Competitive storage and commodity price dynamics. *Journal of Political Economy 104*(5), 896–923.

Dozat, T. (2016). Incorporating Nesterov momentum into Adam. In *International Conference on Learning Representations, Workshop Track*.

Driscoll, J. C. and A. C. Kraay (1998). Consistent covariance matrix estimation with spatially dependent panel data. *The Review of Economics and Statistics 80*(4), 549–560.

Ekeland, I., D. Lautier, and B. Villeneuve (2019). Hedging pressure and speculation in commodity markets. *Economic Theory 68*(1), 83–123.

Erb, C. B. and C. R. Harvey (2006). The strategic and tactical value of commodity futures. *Financial Analysts Journal 62*(2), 69–97.

Fama, E. F. and K. R. French (1988). Business cycles and the behavior of metals prices. *The Journal of Finance 43*(5), 1075–1093.

Gorton, G. B., F. Hayashi, and K. G. Rouwenhorst (2013). The fundamentals of commodity futures returns. *Review of Finance 17*(1), 35–105.

Goyal, A. and N. Jegadeesh (2018). Cross-sectional and time-series tests of return predictability: What is the difference? *The Review of Financial Studies 31*(5), 1784–1824.

Gu, M., W. Kang, D. Lou, and K. Tang (2025). Relative basis and the expected returns of commodity futures. Discussion Paper 942, London School of Economics, Financial Markets Group.

Gu, S., B. Kelly, and D. Xiu (2020). Empirical asset pricing via machine learning. *The Review of Financial Studies 33*(5), 2223–2273.

Hansen, B. E. (1996). Inference when a nuisance parameter is not identified under the null hypothesis. *Econometrica 64*(2), 413–430.

Hansen, B. E. (2017). Regression kink with an unknown threshold. *Journal of Business & Economic Statistics 35*(2), 228–240.

Harvey, C. R., Y. Liu, and H. Zhu (2016). ... and the cross-section of expected returns. *The Review of Financial Studies 29*(1), 5–68.

Hazelkorn, T. M., T. J. Moskowitz, and K. Vasudevan (2023). Beyond basis basics: Liquidity demand and deviations from the law of one price. *The Journal of Finance 78*(1), 301–345.

Hirshleifer, D. (1988). Residual risk, trading costs, and commodity futures risk premia. *The Review of Financial Studies 1*(2), 173–193.

Hirshleifer, D. (1990). Hedging pressure and futures price movements in a general equilibrium model. *Econometrica 58*(2), 411–428.

Holm, S. (1979). A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics 6*(2), 65–70.

Huang, D., J. Li, L. Wang, and G. Zhou (2020). Time series momentum: Is it there? *Journal of Financial Economics 135*(3), 774–794.

Hurst, B., Y. H. Ooi, and L. H. Pedersen (2017). A century of evidence on trend-following investing. *The Journal of Portfolio Management 44*(1), 15–29.

ICE Futures Europe (2024). Commitments of traders reports. Intercontinental Exchange.

Israel, R. and T. J. Moskowitz (2013). The role of shorting, firm size, and time on market anomalies. *Journal of Financial Economics 108*(2), 275–301.

Kaldor, N. (1939). Speculation and economic stability. *The Review of Economic Studies 7*(1), 1–27.

Kang, W., K. G. Rouwenhorst, and K. Tang (2020). A tale of two premiums: The role of hedgers and speculators in commodity futures markets. *The Journal of Finance 75*(1), 377–417.

Keynes, J. M. (1930). *A Treatise on Money, Vol. I: The Pure Theory of Money*. London: Macmillan.

Kim, A. Y., Y. Tse, and J. K. Wald (2016). Time series momentum and volatility scaling. *Journal of Financial Markets 30*, 103–124.

Kogan, L., D. Livdan, and A. Yaron (2009). Oil futures prices in a production economy with investment constraints. *The Journal of Finance 64*(3), 1345–1375.

Koijen, R. S. J., T. J. Moskowitz, L. H. Pedersen, and E. B. Vrugt (2018). Carry. *Journal of Financial Economics 127*(2), 197–225.

Kwon, K. Y., J. Kang, and J. Yun (2021). Basis-momentum strategies and ranking periods. *Finance Research Letters 43*, 101997.

Ledoit, O. and M. Wolf (2008). Robust performance hypothesis testing with the Sharpe ratio. *Journal of Empirical Finance 15*(5), 850–859.

Lim, B., S. Zohren, and S. Roberts (2019). Enhancing time-series momentum strategies using deep neural networks. *The Journal of Financial Data Science 1*(4), 19–38.

London Metal Exchange (2024). Commitments of traders reports. London Metal Exchange.

Loshchilov, I. and F. Hutter (2017). SGDR: Stochastic gradient descent with warm restarts. In *International Conference on Learning Representations*.

McLean, R. D. and J. Pontiff (2016). Does academic research destroy stock return predictability? *The Journal of Finance 71*(1), 5–32.

Moskowitz, T. J., Y. H. Ooi, and L. H. Pedersen (2012). Time series momentum. *Journal of Financial Economics 104*(2), 228–250.

Moskowitz, T. J., R. Sabbatucci, A. Tamoni, and B. Uhl (2025). Nonlinear time series momentum. Working paper, SSRN 5933974.

National Bureau of Economic Research (2024). US business cycle expansions and contractions. https://www.nber.org/research/data/us-business-cycle-expansions-and-contractions.

Newey, W. K. and K. D. West (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica 55*(3), 703–708.

Paschke, R., M. Prokopczuk, and C. Wese Simen (2020). Curve momentum. *Journal of Banking & Finance 113*, 105718.

Petersen, M. A. (2009). Estimating standard errors in finance panel data sets: Comparing approaches. *The Review of Financial Studies 22*(1), 435–480.

Politis, D. N. and J. P. Romano (1994). The stationary bootstrap. *Journal of the American Statistical Association 89*(428), 1303–1313.

Routledge, B. R., D. J. Seppi, and C. S. Spatt (2000). Equilibrium forward curves for commodities. *The Journal of Finance 55*(3), 1297–1338.

Stambaugh, R. F., J. Yu, and Y. Yuan (2012). The short of it: Investor sentiment and anomalies. *Journal of Financial Economics 104*(2), 288–302.

Symeonidis, L., M. Prokopczuk, C. Brooks, and E. Lazar (2012). Futures basis, inventory and commodity price volatility: An empirical analysis. *Economic Modelling 29*(6), 2651–2663.

Szymanowska, M., F. De Roon, T. Nijman, and R. Van Den Goorbergh (2014). An anatomy of commodity futures risk premia. *The Journal of Finance 69*(1), 453–482.

U.S. Commodity Futures Trading Commission (2024). Commitments of traders reports. https://www.cftc.gov/MarketReports/CommitmentsofTraders.

Working, H. (1949). The theory of price of storage. *The American Economic Review 39*(6), 1254–1262.

Appendix

This Online Appendix contains the data description, supporting evidence, and the specification of the neural network cited in the article. Its sections follow the sections of the article, and each section states which claim of the article its exhibits support. Exhibits carry the prefix A. Unless stated otherwise, regressions include market fixed effects and report t-statistics clustered by month, and the signal is $s_{i,t} = BM_{i,t}/\hat{\sigma}_{12}(BM)_{i,t}$ with $s^+ = \max(s, 0)$ and $s^- = \min(s, 0)$, as in Section 3 of the article.

A. Data

This section supports Section 3 of the article.

Exhibit A1 lists the 45 commodity futures contracts with their sector, trading exchange, Bloomberg ticker, first observation month, delivery months, and the source of the trader positions. In this and the following tables, the sector labelled Livestock includes the dairy contracts (butter, cheese, and milk). Basis momentum is available for 43 markets, because propane and pork bellies lack a second-nearby series.

Exhibit A2 shows how the coverage of the dataset grows over time. The number of markets rises from 6 in 1960 to 30 in 1990 and 41 in 2010, and all six sectors are present from 1989. Results that require all sectors, such as the robustness check from 1989 in Exhibit 5 of the article, start in that year.

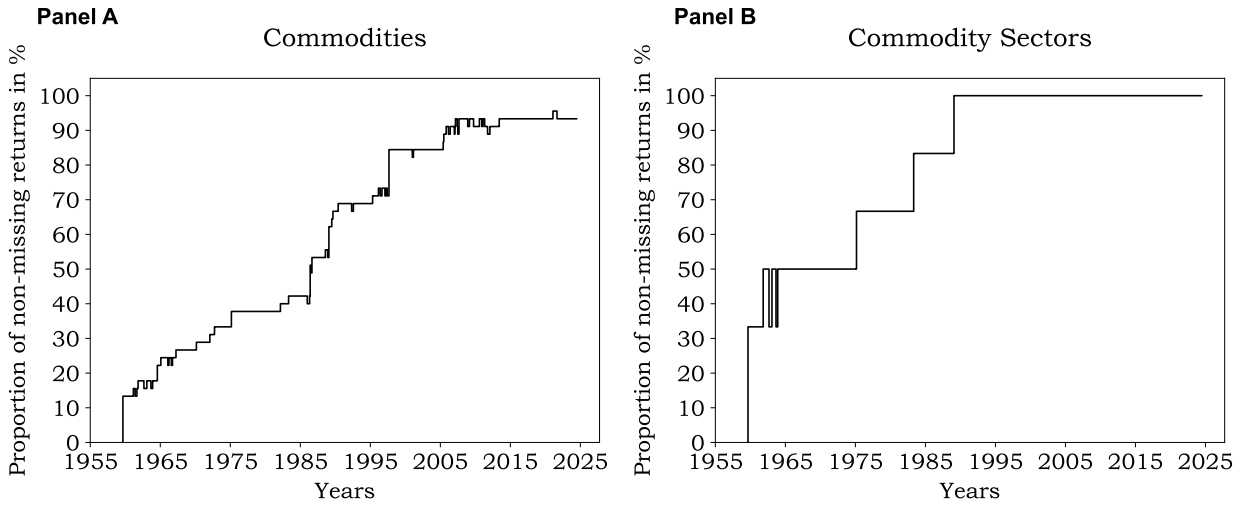


Exhibit A2: **Coverage of the Dataset, August 1959–June 2024.** Panel A shows the share of the 45 commodity futures and Panel B the share of the six sectors with a non-missing monthly return.

Sector	Commodity future	Exchange	Bloomberg Ticker	Start Date	Delivery Months	Positions
Energy	Brent Crude Oil	ICE	CO	1988:06	1:12	ICE COT
	Gas Oil	ICE	QS	1989:07	1:12	ICE COT
	WTI Crude Oil	NYMEX	CL	1983:03	1:12	067651
	Heating Oil	NYMEX	HO	1986:07	1:12	022651
	Natural Gas	NYMEX	NG	1990:04	1:12	023651
	Propane	NYMEX	PN	1988:12	1:12	066651, 06665O
	Unlead. Gasoline	NYMEX	HU/XB	1986:04	1:12	111652, 111659
	Ethanol	CME	DL	2005:05	1:12	025601
Grains & Oilseeds	Canola	ICE	RS	1982:01	1, 3, 5, 7, 11	135731
	Palm Oil	CME	KO	1995:03	1:12	037021
	Oats	CBOT	O	1959:07	3, 5, 7, 9, 12	004601, 004603
	Rough Rice	CBOT	RR	1988:12	1, 3, 5, 7, 9, 11	039781, 039601
	Corn	CBOT	C	1959:07	3, 5, 7, 9, 12	002601, 002602
	Soybean Meal	CBOT	SM	1964:06	1, 3, 5, 7, 8, 9, 10, 12	026603
	Soybean Oil	CBOT	BO	1964:06	1, 3, 5, 7, 8, 9, 10, 12	007601
	Soybeans	CBOT	S	1959:07	1, 3, 5, 7, 8, 9, 11	005601, 005602
	Wheat (SRW)	CBOT	W	1959:07	2, 3, 5, 7, 9, 12	001601, 001602
	Wheat (HRW)	CBOT	KW	1970:01	3, 5, 7, 9, 12	001611, 001612
	Barley	NDX	FU	2006:12	4, 5, 6, 7	NA
	Cottonseed	NDX	C7	2005:04	4, 5, 6, 7	NA
Livestock	Butter	CME	V6	2005:09	1:12	050642
	Cheese	CME	CHE	2010:06	1:12	063642
	Feeder Cattle	CME	FC	1971:12	1, 4, 5, 8, 9, 11	061641
	Lean Hogs	CME	LH	1986:04	2, 4, 5, 6, 7, 8, 10, 12	054641, 054642
	Live Cattle	CME	LC	1964:12	2, 4, 6, 8, 10, 12	057642
	Pork Bellies	CME	PB	1961:09	2, 3, 10, 11, 12	056641
	Milk	CME	DA	1996:01	1:12	052641
Industrial Metals	Aluminium	LME	LA	1997:07	1:12	LME COTR
	Cobalt	CME	CVT	2020:12	1:12	188691
	Copper	COMEX	HG	1988:12	3, 5, 7, 9, 12	085691, 085692
	Iron Ore	SGX	SCO	2013:04	1:12	195651
	Lead	LME	LL	1997:07	1:12	LME COTR
	Nickel	LME	LN	1997:07	1:12	LME COTR
	Tin	LME	LT	1997:07	1:12	LME COTR
	Zinc	LME	LX	1997:07	1:12	LME COTR
Precious Metals	Gold	COMEX	GC	1975:01	2, 4, 6, 8, 10, 12	088691
	Palladium	NYMEX	PA	1986:04	3, 6, 9, 12	075651
	Platinum	NYMEX	PL	1986:04	1, 4, 7, 10	076651
	Silver	COMEX	SI	1975:01	3, 5, 7, 9, 12	084691
Softs	Cocoa	ICE	CC	1959:07	3, 5, 7, 9, 12	073732
	Coffee	ICE	KC	1972:08	3, 5, 7, 9, 12	083731
	Cotton	ICE	CT	1959:07	3, 5, 7, 10, 12	033661
	Orange Juice	ICE	JO	1967:02	1, 3, 5, 7, 9, 11	040701
	Sugar	ICE	SB	1961:01	3, 5, 7, 10	080732
	Rubber	TOCOM	JN	1989:05	1:12	NA

Exhibit A1: **Commodity Futures.** For each of the 45 commodity futures, we report the sector, the exchange on which it is traded, the Bloomberg ticker, the month of the first recorded settlement price, the delivery months, and the source of the trader positions: the code in the Commitments of Traders reports of the Commodity Futures Trading Commission (CFTC), or the reports of ICE and the LME. The six sectors and their sizes are Energy (8), Grains & Oilseeds (12), Livestock (7), Industrial Metals (8), Precious Metals (4), Softs (6). Unleaded Gasoline uses the HU contract until September 2006 and the XB contract thereafter, spliced on returns; because Bloomberg's adjusted HU series are not roll-adjusted, HU returns are computed from unadjusted prices on the NYMEX roll calendar. Lumber is excluded because its Bloomberg series are not roll-adjusted and its roll dates cannot be recovered. Returns are available through June 2024 except for Propane (August 2009), Ethanol (July 2021), Pork Bellies (March 2011). Basis momentum requires a second-nearby contract and is therefore not available for Propane and Pork Bellies. Positions for Brent Crude Oil and Gas Oil are producer/merchant positions from the ICE reports (from 2011), and for the LME metals commercial positions from the LME reports (from 2020). NA: no positions available.

Exhibit A3 reports descriptive statistics of the monthly excess returns of the first-nearby contracts. Average returns differ widely across markets, from -16.2% per year for natural gas to 34.0% for ethanol, and annualized volatilities range from about 16% for cattle and dairy to 64% for propane. These differences motivate the scaling of returns by their ex-ante volatility and the market fixed effects in all regressions. Monthly returns are close to serially uncorrelated; the first-order autocorrelation exceeds 0.15 only for three energy contracts, the three dairy contracts, and cobalt.

Sector	Commodity	# of obs.	Mean (%) p.a.	Volatility (%) p.a.	Skewness p.a.	Kurtosis p.a.	Autocorr. $\rho(1)$
Energy	Brent Crude Oil	432	9.789	34.785	0.022	3.367	0.206
	Gas Oil	419	8.463	33.417	0.070	3.120	0.194
	WTI Crude Oil	495	3.950	38.679	0.287	3.878	0.152
	Heating Oil	455	10.089	36.430	0.262	3.362	0.091
	Natural Gas	410	−16.208	58.849	0.308	3.298	0.010
	Propane	248	16.216	63.928	1.935	9.509	−0.052
	Unlead. Gasoline	456	14.700	35.378	−0.042	3.372	0.113
	Ethanol	194	34.029	37.627	0.283	3.258	0.117
Grains & Oilseeds	Canola	509	−1.016	22.065	0.054	3.191	0.024
	Palm Oil	351	3.213	32.554	0.171	3.183	0.084
	Oats	772	−2.742	30.646	0.551	4.391	−0.017
	Rough Rice	423	−6.639	25.677	0.236	3.377	−0.016
	Corn	779	−4.299	24.155	0.329	3.521	0.014
	Soybean Meal	720	6.862	31.682	0.608	4.311	0.049
	Soybean Oil	720	4.377	30.881	0.410	3.524	−0.020
	Soybeans	779	2.699	25.498	0.391	3.832	0.016
	Wheat (SRW)	779	−5.886	25.775	0.186	3.239	0.036
	Wheat (HRW)	653	−1.121	26.304	0.225	3.278	0.081
	Barley	208	1.077	26.845	0.552	3.859	0.121
	Cottonseed	218	3.718	26.657	0.044	3.023	−0.007
Livestock	Butter	222	−1.587	16.072	−0.105	3.382	0.230
	Cheese	168	2.309	16.664	0.341	3.437	0.171
	Feeder Cattle	630	1.052	16.630	−0.116	3.180	−0.019
	Lean Hogs	458	−1.533	28.566	0.081	3.161	−0.033
	Live Cattle	714	3.460	16.770	−0.038	3.140	0.027
	Pork Bellies	584	−2.757	35.852	0.129	3.124	−0.071
	Milk	336	0.695	17.243	0.017	3.375	0.178
Industrial Metals	Aluminium	323	−2.958	19.756	0.053	3.007	0.049
	Cobalt	42	−14.764	35.122	0.242	3.077	0.296
	Copper	426	4.202	24.839	0.037	3.236	0.066
	Iron Ore	134	22.551	38.134	−0.020	2.974	0.098
	Lead	321	3.009	27.313	0.026	3.101	−0.001
	Nickel	323	3.261	34.787	0.086	3.027	0.065
	Tin	323	8.149	25.364	0.104	3.127	0.136
	Zinc	323	−0.394	26.494	−0.004	3.099	−0.011
Precious Metals	Gold	593	0.375	18.473	0.122	3.282	−0.010
	Palladium	458	4.185	31.719	0.099	3.228	−0.002
	Platinum	458	1.188	21.886	−0.010	3.257	0.007
	Silver	593	−1.438	31.681	0.158	3.428	0.050
Softs	Cocoa	776	2.200	31.800	0.278	3.291	0.001
	Coffee	620	0.061	36.282	0.306	3.263	−0.022
	Cotton	775	0.294	24.896	0.128	3.214	0.043
	Orange Juice	688	3.162	32.903	0.436	3.660	−0.060
	Sugar	761	−6.739	42.219	0.317	3.286	0.141
	Rubber	421	−2.448	34.138	0.134	3.105	0.073

Exhibit A3: **Commodity Futures Descriptive Statistics.** For each commodity future we report the number of monthly observations, the annualized geometric average return, the annualized standard deviation, the skewness, the kurtosis, and the first-order autocorrelation of monthly excess returns from August 1959 to June 2024. Skewness and kurtosis are annualized as for a sum of twelve independent monthly returns: the monthly skewness is divided by $\sqrt{12}$, and the kurtosis is three plus the monthly excess kurtosis divided by 12. The autocorrelation pairs consecutive months only.

B. The Hockey Stick

This section supports Section 4 of the article: the shape of the relation, the location of the kink, and its stability across sectors, markets, and controls.

Exhibit A4 documents the linear time-series strategy that the article takes as its benchmark (Section 4.4 of the article). Panel A shows that its Sharpe ratio is 0.71 over 1960–2024 but −0.03 in 2000–2009. Panel B shows that the strategy has a beta of −0.39 on the long-only portfolio, because it is short more often than long. Panel C shows that the failure in the 2000s is not only a matter of this exposure: the linear predictive slope, estimated with market fixed effects, falls to 0.005 (t-statistic of 0.4) in 2000–2009. Panel D attributes the failure to the positions taken on negative signals, which earn a Sharpe ratio of 0.06 over the full sample and −0.73 in 2000–2009, whereas the positions on positive signals are profitable in every subperiod.

Exhibit A5 repeats Exhibit 1 of the article with ten bins, separately for 1960–1999 and 2000–2024 (Section 4.1 of the article). It shows that the shape is not driven by one half of the sample: in both periods, the highest decile of the signal is followed by returns significantly above the market mean, and the deciles of negative signals show no steady decline.

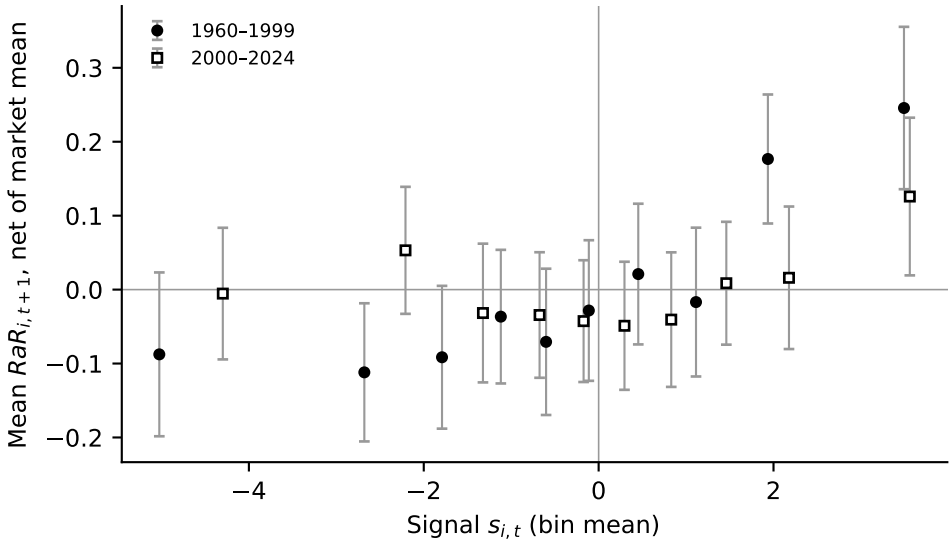


Exhibit A5: **The Shape of the Relation before and after 2000.** Mean of $RaR_{i,t+1}$ net of each market's mean in the period, in ten pooled quantile bins of the signal $s_{i,t}$, separately for 1960–1999 and 2000–2024, with 95% confidence intervals (standard errors clustered by month).

Exhibit A4: **Linear Time-Series Basis Momentum.**

	1960–2024	1960–1989	1990–1999	2000–2009	2010–2024
Panel A: Sharpe ratio of the linear basis-momentum strategy					
All markets	0.71***	0.82***	1.20***	−0.03	0.75***
Energy	0.46***	0.66	0.97***	0.12	0.33
Grains & Oilseeds	0.48***	0.60***	0.34	0.15	0.49**
Industrial Metals	0.18		0.44	−0.27	0.48*
Livestock	0.39***	0.67***	0.33	0.27	−0.25
Precious Metals	0.00	−0.35	0.46*	−0.30	0.38*
Softs	0.35***	0.35**	0.37*	0.07	0.72**
Long-only, all markets	0.41	0.41	−0.04	0.82	0.40
Panel B: Alpha of the strategy on the long-only portfolio					
α (% per month)	2.37***	3.68***	2.33***	0.39	1.10**
	(6.12)	(6.01)	(3.74)	(0.49)	(2.57)
β long-only	−0.39	−0.63	−0.14	−0.32	0.32
Appraisal ratio	0.83	1.04	1.20	0.20	0.66
Panel C: Predictive panel regression $RaR_{i,t+1} = a_i + b\, s_{i,t} + \varepsilon_{i,t+1}$					
b	0.026***	0.040***	0.036***	0.005	0.018**
	(5.08)	(4.40)	(2.75)	(0.41)	(2.10)
Markets	43	23	36	40	43
Observations	19,336	4,500	3,367	4,308	7,161
Panel D: Sharpe ratio of the positions with positive and negative signals					
Positions with $s_{i,t} > 0$	0.93***	1.04***	0.90***	1.05***	0.71**
Positions with $s_{i,t} < 0$	0.06	0.15	0.60*	−0.73	0.01
Share of positions long	0.45	0.42	0.46	0.41	0.55

The strategy holds in each market a position $w_{i,t} = s_{i,t}$ with $s_{i,t} = BM_{i,t}/\hat{\sigma}_{12}(BM)_{i,t}$ and earns $w_{i,t}\,\sigma^{target} RaR_{i,t+1}$, with $RaR_{i,t+1} = r_{i,t+1}/\hat{\sigma}_{12}(r)_{i,t}$ and $\sigma^{target} = 40\%$ per year; portfolios are equally weighted across the markets with a signal. Panel A: annualized Sharpe ratios; stars refer to one-sided stationary-bootstrap p-values (months resampled, mean block length 6, 2,500 draws) of H_0: Sharpe ≤ 0. The long-only portfolio holds the same markets with $w_{i,t} = 1$. Panel B: OLS of the all-market strategy return on the long-only return, Newey–West t-statistics (6 lags) in parentheses; the appraisal ratio is the annualized α over the residual volatility. Panel C: market fixed effects, standard errors clustered by month, t-statistics in parentheses; a month bootstrap of b over 1960–2024 gives $p = 0.000$. Panel D splits the strategy return exactly into the contributions of the positions with $s_{i,t} > 0$ and $s_{i,t} < 0$ (each divided by the number of markets in the portfolio), with stars as in Panel A. ***, **, * denote significance at the 1%, 5%, and 10% levels.

Exhibit A6 illustrates the regression kink of Exhibit 3 of the article (Section 4.3). Panel A plots the F-statistic of the kink model against the linear model for each candidate location. The profile peaks at $\hat{\gamma} = -0.10$ and falls steeply for locations above one standard deviation, but it declines only slowly toward negative locations: a kink at -3 still yields about half of the maximum F-statistic. This is why the 90% interval of the location extends far into negative signals although the test clearly rejects a linear relation. Panel B overlays the fitted kink on the binned means.

Exhibit A7 estimates the asymmetric regression of Exhibit 2 of the article separately for each sector (Section 4.4 of the article). The slope on positive basis momentum is positive in all six sectors and significant in five of them, and the slope on negative basis momentum

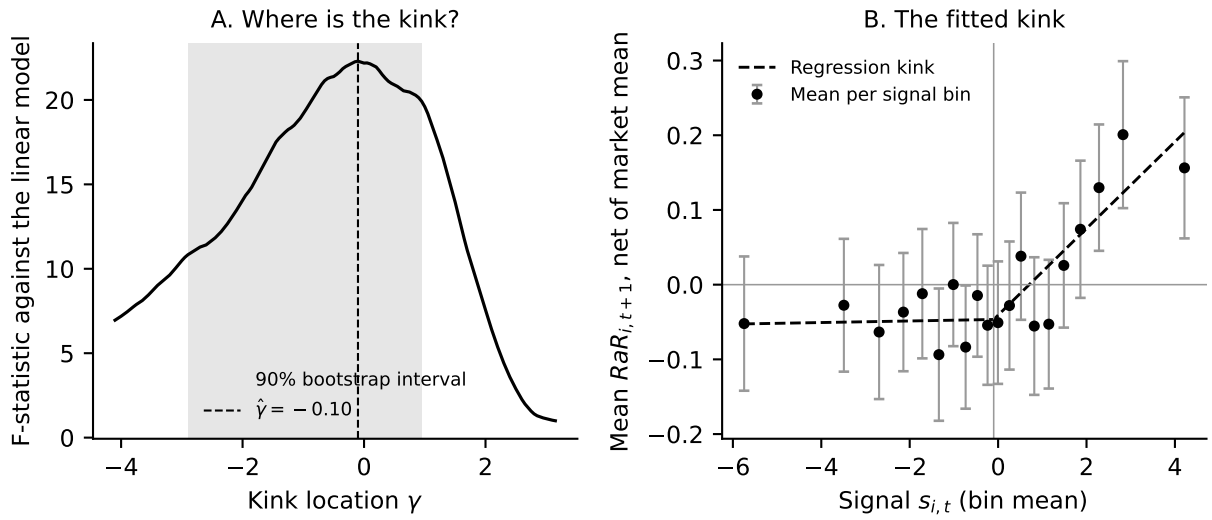


Exhibit A6: **Where Is the Kink?** Panel A: F-statistic of the regression kink of Equation (7) against the linear model as a function of the kink location γ, 1960–2024, with the estimate $\hat{\gamma}$ and its 90% bootstrap interval. Panel B: mean of $RaR_{i,t+1}$ net of each market's mean in 20 quantile bins of $s_{i,t}$ with the fitted regression kink.

is insignificant in all six. Within a single sector, the difference between the two slopes is estimated less precisely: it is significant at the 5% level in grains and oilseeds and in industrial metals, and after a Holm adjustment for six tests, no sector difference is significant at the 5% level. The sector results therefore show that the pattern is common to the sectors, not that each sector establishes it on its own.

Exhibit A7: **Asymmetry by Sector.**

Sector	Markets	b^+	b^-	$t(b^+ - b^-)$	p	p_{Holm}
Energy	7	0.064**	−0.004	1.58	0.114	0.278
		(2.43)	(−0.16)			
Grains & Oilseeds	12	0.060***	0.009	2.27	0.023	0.116
		(3.67)	(0.79)			
Industrial Metals	8	0.080**	−0.038	2.54	0.011	0.066
		(2.57)	(−1.52)			
Livestock	6	0.037	0.002	0.92	0.360	0.360
		(1.61)	(0.10)			
Precious Metals	4	0.063**	−0.021	1.95	0.051	0.203
		(2.38)	(−0.82)			
Softs	6	0.061***	0.013	1.68	0.093	0.278
		(3.03)	(0.88)			

Asymmetric regression $RaR_{i,t+1} = a_i + b^+ s^+_{i,t} + b^- s^-_{i,t} + \varepsilon_{i,t+1}$, $s^+ = \max(s, 0)$, $s^- = \min(s, 0)$, estimated separately for each sector, 1960–2024, market fixed effects, standard errors clustered by month (t-statistics in parentheses). $t(b^+ - b^-)$ and the Wald p-value test $b^+ = b^-$; the last column adjusts the p-values across the six sectors (Holm, 1979).

Exhibit A8 addresses two concerns about inference. Panel A asks whether the t-statistic

of the difference could arise under a linear relation, given the cross-sectional correlation and the fat tails of commodity returns. A wild bootstrap that imposes a linear relation gives a 99% critical value of 2.22, far below the observed 4.17. Panel B asks whether a few markets drive the pooled result. Estimated market by market, the slope on positive basis momentum is positive in 33 of 42 markets and exceeds that on negative basis momentum in 32; the slope on negative basis momentum is significant in only 5 markets.

Exhibit A8: **Inference Checks for the Asymmetry.**

Panel A: Null-imposed bootstrap (linear relation under H_0)		
Observed $t(b^+ - b^-)$	4.17	
Bootstrap critical value, 95% / 99%	1.62 / 2.22	
Bootstrap p-value	<0.001	
Panel B: Market-by-market regressions (42 markets)		
	Number of markets	Binomial p
$b_i^+ > 0$	33	0.0001
$b_i^+ > b_i^-$	32	0.0005
$t(b_i^+) > 1.65$	14	0.0000
$t(b_i^-) > 1.65$	5	0.0573
Median b_i^+ / median b_i^-	0.054 / −0.007	

Panel A: under the null of a linear relation, $RaR^*_{i,t+1} = \hat{a}_i + \hat{b}s_{i,t} + \hat{e}_{i,t+1}v_{t+1}$ with one standard normal draw v_{t+1} per month (wild bootstrap by month, 999 draws); the asymmetric regression with market fixed effects is re-estimated on each sample (Huang et al., 2020). Panel B: for each market with at least 60 observations, OLS of $RaR_{i,t+1}$ on $s^+_{i,t}$ and $s^-_{i,t}$ with Newey–West t-statistics (6 lags). Binomial p-values test whether the count exceeds what is expected by chance (one half for signs, 5% for one-sided significance).

Exhibit A9 asks whether basis momentum proxies for momentum, the basis, or hedging pressure (Section 4.5 of the article). Panel A adds the positive and negative parts of each control, in their common sample from 1986. The slope on positive basis momentum falls at most from 0.055 to 0.044, the slope on negative basis momentum stays at zero, and the difference remains significant in every column (t-statistics between 2.9 and 3.6). Panel B sorts on the sign of the signal within terciles of each control. In all nine terciles, returns after positive signals exceed those after negative signals and the positive-side slope exceeds the negative-side slope; the difference in slopes is significant at the 5% level in four terciles.

Exhibit A9: **Basis Momentum, Momentum, the Basis, and Hedging Pressure.**

Panel A: Asymmetric regressions with controls

	BM only	+ Momentum	+ Basis	+ HP	+ All
b^+	0.055***	0.044***	0.055***	0.052***	0.048***
	(4.79)	(3.87)	(4.80)	(4.65)	(4.16)
b^-	−0.005	−0.005	−0.005	−0.003	−0.003
	(−0.46)	(−0.50)	(−0.55)	(−0.35)	(−0.35)
$t(b^+ - b^-)$	3.53	2.94	3.60	3.35	3.05
Observations	12,283	12,283	12,283	12,283	12,283

Panel B: Double sorts on the sign of s and terciles of the control

		Mean $RaR_{i,t+1}$, net of market mean			Regression within tercile		
Control	Tercile	$s > 0$	$s < 0$	Difference	b^+	b^-	$t(b^+ - b^-)$
Momentum	Low	−0.020	−0.068	0.048 (1.29)	0.020	0.014	0.24
	Mid	0.038	−0.067	0.105*** (3.73)	0.080***	−0.008	3.63
	High	0.095	0.032	0.063* (1.81)	0.054***	−0.010	2.65
Basis	Low	0.017	−0.067	0.083** (2.49)	0.048**	0.007	1.68
	Mid	0.019	−0.030	0.049 (1.54)	0.050***	0.007	1.71
	High	0.088	−0.030	0.118*** (3.47)	0.071***	−0.036*	3.90
Hedging pressure	Low	0.027	−0.027	0.054 (1.27)	0.039*	−0.004	1.47
	Mid	0.007	−0.029	0.035 (0.88)	0.045**	−0.005	1.65
	High	0.051	−0.035	0.086** (2.30)	0.063***	−0.003	2.50

Panel A: $RaR_{i,t+1} = a_i + b^+ s^+_{i,t} + b^- s^-_{i,t} + c^+ x^+_{i,t} + c^- x^-_{i,t} + \varepsilon_{i,t+1}$, where x is 12-1 month momentum divided by $\hat{\sigma}_{12}(r)_{i,t}$, the basis $\ln F^{(1)}_{i,t} - \ln F^{(2)}_{i,t}$ from unadjusted prices divided by its expanding standard deviation, or the hedging pressure of commercial traders (Exhibit 8); all columns use the sample in which all three controls are available (1986–2024). Panel B: each month, markets are sorted into terciles of the control; within each tercile, the mean next-month risk-adjusted return net of the market mean for positive and negative signals, their difference, and the asymmetric regression (full sample of each control). Pooled correlations of $s_{i,t}$ with momentum, the basis, and hedging pressure: 0.21, 0.32, and −0.02. Market fixed effects; t-statistics clustered by month in parentheses; ***, **, * denote significance at the 1%, 5%, and 10% levels.

C. Which Information Does Each Side Carry?

This section supports Section 5 of the article: the relative information of negative basis momentum, the tests of differences across groups and states, and the persistence of the relation.

Exhibit A10 splits the cross-sectional basis-momentum factor of Boons and Porras Prado (2019) into a long leg (top tercile minus the average market) and a short leg (average market minus bottom tercile) (Section 5.1 of the article). Over 1960–2024, both legs earn significant alphas on the average market (0.56% and 0.40% per month). Markets with the lowest basis momentum thus underperform other markets even though, in the time series, negative basis momentum does not predict falling prices. This is the relative information that the article attributes to the negative side. After 2000, the alpha of the short leg is insignificant (0.12%, t-statistic of 1.2).

Exhibit A11 reports the formal tests of differences that Sections 5.2 and 5.3 of the article interpret. Panel A estimates group-specific slopes in one regression. The positive-side slope

Exhibit A10: **The Asymmetry in the Cross-Sectional Basis-Momentum Factor.**

	1960–2024			1960–1999			2000–2024		
	Mean	SR	α	Mean	SR	α	Mean	SR	α
Long-short factor	0.99	0.74***	0.97***	1.36	0.90***	1.37***	0.47	0.45***	0.38*
			(5.86)			(6.15)			(1.82)
Long leg (top minus average)	0.56	0.76***	0.56***	0.76	0.96***	0.78***	0.28	0.44**	0.25*
			(6.15)			(6.63)			(1.96)
Short leg (average minus bottom)	0.43	0.54***	0.40***	0.61	0.65***	0.59***	0.19	0.34**	0.12
			(4.17)			(4.22)			(1.18)

Each month, markets with a signal (at least nine) are sorted into terciles of $BM_{i,t}$; equal-weighted excess returns of month $t+1$. The long-short factor (Boons and Porras Prado, 2019) equals the long leg (top tercile minus the equal-weighted average of all markets) plus the short leg (average minus bottom tercile). Mean and α (on the equal-weighted average) in % per month; Sharpe ratios annualized, stars refer to one-sided bootstrap p-values; Newey–West t-statistics in parentheses.

is the same in storable commodities and precious metals (0.063 each), and none of the differences between the groups is significant; precious metals, livestock, and dairy taken together show a significant asymmetry of their own (t-statistic of 2.0). Panel B replaces the state indicators of Exhibit 8 of the article by continuous interactions. The interaction with the standardized basis is zero (t-statistic of -0.4), and that with hedging pressure is positive and marginally significant (1.8). Panel C estimates the kink separately in each state. The locations order as scarcity implies, but the bootstrap intervals of the shifts include zero (p-values of 0.15 and 0.30).

Exhibit A12 asks whether the positive-side slope merely reflects bad times (Section 5.3 of the article). In NBER recessions and in months of high commodity volatility, the slope on positive basis momentum is similar to that outside these states (Wald p-values of 0.77 and 0.90). In recessions, which cover 10% of the months, the slope on negative basis momentum is positive and marginally significant (0.057, t-statistic of 1.9), so that the asymmetry is weaker in these few months.

Exhibit A11: **Differences across Commodity Groups and States.**

Panel A: Group-specific slopes in one regression

	b^+	b^-	$b^+ - b^-$
Storable	0.063***	0.004	0.059***
	(5.83)	(0.46)	(3.76)
Precious metals	0.063**	−0.021	0.084*
	(2.38)	(−0.82)	(1.95)
Livestock and dairy	0.037	0.002	0.035
	(1.62)	(0.10)	(0.92)
Precious metals, livestock, and dairy			0.059**
			(2.04)

	Difference in b^+	Difference in $b^+ - b^-$
Storable − precious metals	0.000 (0.00)	−0.025 (−0.55)
Storable − livestock and dairy	0.025 (1.01)	0.024 (0.58)
Storable − all other markets	0.013 (0.64)	0.000 (−0.01)

Panel B: Continuous interactions with the state variable $z_{i,t}$

	Basis (standardized)	Hedging pressure
s^+	0.061***	0.039***
	(6.26)	(2.85)
s^-	−0.005	0.001
	(−0.53)	(0.13)
$s^+ \times z$	−0.003	0.088*
	(−0.42)	(1.76)
$s^- \times z$	−0.009	−0.045
	(−1.07)	(−1.01)
Mean (standard deviation) of z	−0.18 (1.03)	0.12 (0.23)

Panel C: Location of the kink by state

	$\hat{\gamma}$ in first state	$\hat{\gamma}$ in second state	Shift [90% interval], p
Contango vs. backwardation	1.05	−0.75	1.80 [−2.65, 4.15], 0.148
Hedgers net long vs. net short	1.35	−1.30	2.65 [−4.12, 3.71], 0.297

Panel A: $RaR_{i,t+1} = a_i + \sum_g \left(b_g^+ s_{i,t}^+ 1_{i\in g} + b_g^- s_{i,t}^- 1_{i\in g}\right) + \varepsilon_{i,t+1}$ with g = storable commodities (energy, grains and oilseeds, industrial metals, softs), precious metals, and livestock and dairy, 1960–2024; the row for all other markets and the comparison with them come from a regression with two groups. Panel B: $RaR_{i,t+1} = a_i + b^+ s_{i,t}^+ + b^- s_{i,t}^- + c^+ s_{i,t}^+ z_{i,t} + c^- s_{i,t}^- z_{i,t} + d\, z_{i,t} + \varepsilon_{i,t+1}$, where z is the basis $\ln F^{(1)} - \ln F^{(2)}$ divided by its expanding standard deviation or the hedging pressure of commercial traders. Market fixed effects; t-statistics clustered by month in parentheses. Panel C: regression kink of Equation (7) estimated separately in the two states (Exhibit 8); shift = first minus second location, with the 90% interval and the one-sided p-value of H_0: shift ≤ 0 from a stationary month-block bootstrap (mean block length 6, 499 draws) that re-estimates both locations. ***, **, * denote significance at the 1%, 5%, and 10% levels.

Exhibit A13 reports the slopes of regressions of cumulative returns over the next h months on the positive and negative parts of the signal, divided by h. The slope on positive basis momentum declines only gradually, from 0.059 at one month to 0.035 per month at twelve months, and the slope on negative basis momentum is insignificant at every horizon. Because the horizons overlap, the confidence intervals use Driscoll and Kraay (1998) standard errors.

Exhibit A12: **State Dependence of the Asymmetry.**

	In state	Not in state	p (equal)
NBER recessions (10% of months)			
b^+	0.045	0.060***	0.769
	(0.90)	(6.12)	
b^-	0.057*	−0.003	0.063
	(1.88)	(−0.34)	
High commodity volatility (26% of months)			
b^+	0.061***	0.058***	0.899
	(3.15)	(4.85)	
b^-	0.015	−0.005	0.282
	(1.02)	(−0.47)	

Asymmetric regression with slopes interacted with a state indicator measured in the month in which the position is formed: NBER recession months, and months in which the trailing 12-month volatility of the long-only commodity portfolio exceeds its expanding median. Market fixed effects; standard errors clustered by month (t-statistics in parentheses); the last column is the p-value of a Wald test of equal slopes in and out of the state.

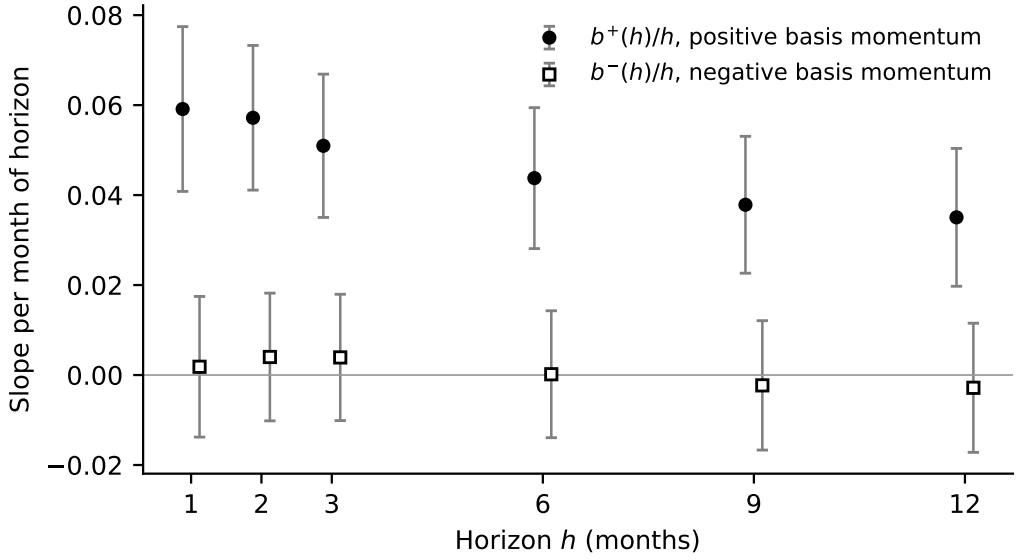


Exhibit A13: **Persistence of the Predictive Relation.** Slopes $b^+(h)$ and $b^-(h)$ of regressions of $\sum_{k=1}^{h} RaR_{i,t+k}$ on $s^+_{i,t}$ and $s^-_{i,t}$ with market fixed effects, divided by h, 1960–2024, with 95% confidence intervals (Driscoll–Kraay standard errors with bandwidth h).

D. Economic Value and Implementability

This section supports Section 6 of the article and three robustness statements of Sections 4.4 and 5.2. All strategies are the walk-forward positions of Exhibit 9 of the article, February

1990–June 2024.

Exhibit A14 collects five checks motivated by the literature. Panel A applies the decomposition of Goyal and Jegadeesh (2018): each strategy return splits exactly into a cross-sectional part, a static net long part, and a timing part. Of the gain of the asymmetric position over linear basis momentum (0.89 percentage points per month), 0.61 comes from the static net long part, which the long-only portfolio spans; the cross-sectional and timing parts add 0.15 and 0.13 (t-statistics of 1.7 and 2.0). The positive-only rule gains only through its net long position; its cross-sectional part is lower than that of linear basis momentum (-0.53, t-statistic of -2.7). Panel B tests the Sharpe-ratio differences with the heteroskedasticity- and autocorrelation-robust method of Ledoit and Wolf (2008). The one-sided p-values of the asymmetric position (0.066 in the walk-forward period, 0.019 after 2000) are close to the bootstrap p-values of the article. Panel C splits the sample at April 2015, when basis momentum first circulated as a working paper. The asymmetry is already absent from 2010 to April 2015, and the change after April 2015 is insignificant, so the weakening is not a publication effect. Panel D shows that the asymmetry comes from months $t-11$ to $t-1$ of the 12-month window, not from the most recent month. Panel E interacts the positive-side slope with a high-VIX state and with backwardation: the VIX interaction is insignificant, and the backwardation interaction remains significant when both are included.

Exhibit A15 reports the walk-forward Sharpe ratios within each sector. The asymmetric position earns a higher Sharpe ratio than linear basis momentum in all six sectors, so the gain of the pooled strategy does not come from a single sector. Exhibit A16 plots the cumulative returns of the walk-forward strategies at a common volatility.

Exhibit A14: **Net Long Exposure, Sharpe-Ratio Tests, Publication, Sub-Windows, and Intermediary Risk.**

Panel A: Decomposition of walk-forward returns, % per month (Goyal and Jegadeesh, 2018)

	Linear BM	Asymmetric	s^+ only	Asym. − linear	s^+ only − linear
Total	1.210***	2.098***	1.397***	0.888***	0.187
	(2.97)	(3.87)	(3.82)	(2.58)	(0.47)
Cross-sectional	1.245***	1.394***	0.714***	0.148*	−0.531***
	(4.12)	(4.58)	(4.71)	(1.73)	(−2.73)
Static net long	−0.109**	0.497**	0.530**	0.606**	0.639**
	(−1.96)	(1.96)	(1.96)	(1.96)	(1.96)
Timing	0.074	0.208	0.153	0.134**	0.079
	(0.36)	(0.87)	(1.45)	(1.98)	(0.61)
Average net position	−0.16	0.72	0.77		

Panel B: Sharpe-ratio difference to linear BM (Ledoit and Wolf, 2008)

	Δ Sharpe	HAC s.e.	p, two-sided	p, one-sided	Bootstrap p
Asymmetric, walk-forward	0.23	0.16	0.132	0.066	0.083
s^+ only, walk-forward	0.22	0.24	0.358	0.179	0.202
Asymmetric, after 2000	0.37	0.18	0.038	0.019	0.024
s^+ only, after 2000	0.43	0.28	0.122	0.061	0.066

Panel C: Before and after the first circulation of basis momentum (April 2015)

	b^+	b^-	$b^+ - b^-$	Observations
1960–Apr 2015	0.068***	−0.003	0.071*** (4.49)	14732
May 2015–2024	0.034*	0.017	0.018 (0.63)	4604
2010–Apr 2015	0.010	0.020	−0.010 (−0.28)	2557
Change after April 2015			−0.028 (−0.79)	

Panel D: Sub-windows of the 12-month window (Kwon, Kang, and Yun, 2021)

	b^+ alone	b^- alone	$t(b^+ - b^-)$ alone	$t(b^+ - b^-)$ joint
Month t	0.026	0.018	0.29	0.29
Months $t-5$ to $t-1$	0.081***	0.006	3.45	3.52
Months $t-11$ to $t-6$	0.051***	0.006	2.22	2.24

Panel E: High VIX and backwardation, 1989–2024 (Cheng et al., 2015)

	s^+	s^-	$s^+ \times$ high VIX	$s^+ \times$ backw.
High VIX	0.041*** (2.68)	−0.010 (−0.84)	0.029 (1.19)	
Backwardation	0.037*** (2.76)	0.002 (0.21)		0.034** (2.05)
Both	0.021 (1.23)	−0.005 (−0.40)	0.030 (1.22)	0.035** (2.10)

Panel A: each walk-forward strategy return (February 1990–June 2024, Exhibit 9) $r_t = \frac{1}{N}\sum_i w_{i,t-1}R_{i,t}$ splits exactly into a cross-sectional part $\frac{1}{N}\sum_i(w_{i,t-1} - \bar{w}_{t-1})(R_{i,t} - \bar{R}_t)$ and a net-exposure part $\bar{w}_{t-1}\bar{R}_t$, which splits into a static part $E[\bar{w}]\bar{R}_t$ and a timing part $(\bar{w}_{t-1} - E[\bar{w}])\bar{R}_t$; mean in % per month with Newey–West t-statistics (6 lags). Panel B: annualized difference of Sharpe ratios to linear basis momentum, standard error from the delta method with a Bartlett-kernel HAC covariance (6 lags), and the one-sided stationary-bootstrap p-value of Exhibit 9. Panel C: asymmetric regression of Exhibit 2 by period; the last row interacts s^+ and s^- with an indicator for months after April 2015 (first SSRN version of Boons and Porras Prado, 2019). Panel D: basis momentum computed over the stated months only, each standardized by its trailing 12-month volatility; alone and jointly in one regression. Panel E: high VIX = month-end VIX (VXO before 1990) above its trailing 60-month median at t; backwardation as in Exhibit 8. Market fixed effects and t-statistics clustered by month in Panels C–E. ***, **, * denote significance at the 1%, 5%, and 10% levels.

Exhibit A17 reports three uses for investors (Section 6.3 of the article). Panel A tilts the long-only portfolio toward markets with high basis momentum without ever shorting. All tilts raise the Sharpe ratio from 0.43 to between 0.76 and 0.79; the asymmetric tilt does not beat the linear one, because the long-only constraint already removes the short side. Panel B adds each position to a commodity multifactor portfolio. The asymmetric position

Exhibit A15: **Out-of-Sample Sharpe Ratios by Sector.**

Sector	Linear BM	Asymmetric	Neural net	Long-only	TSMOM
Energy	0.47	0.50	0.47	0.32	0.31
Grains & Oilseeds	0.34	0.52	0.44	0.23	0.31
Industrial Metals	0.18	0.40	0.30	0.21	0.13
Livestock	0.08	0.18	0.11	0.16	0.16
Precious Metals	0.16	0.41	0.44	0.31	0.41
Softs	0.43	0.47	0.39	0.26	0.45

Annualized Sharpe ratios of the walk-forward strategies of Exhibit 9 of the article (February 1990–June 2024), equal-weighted within each sector.

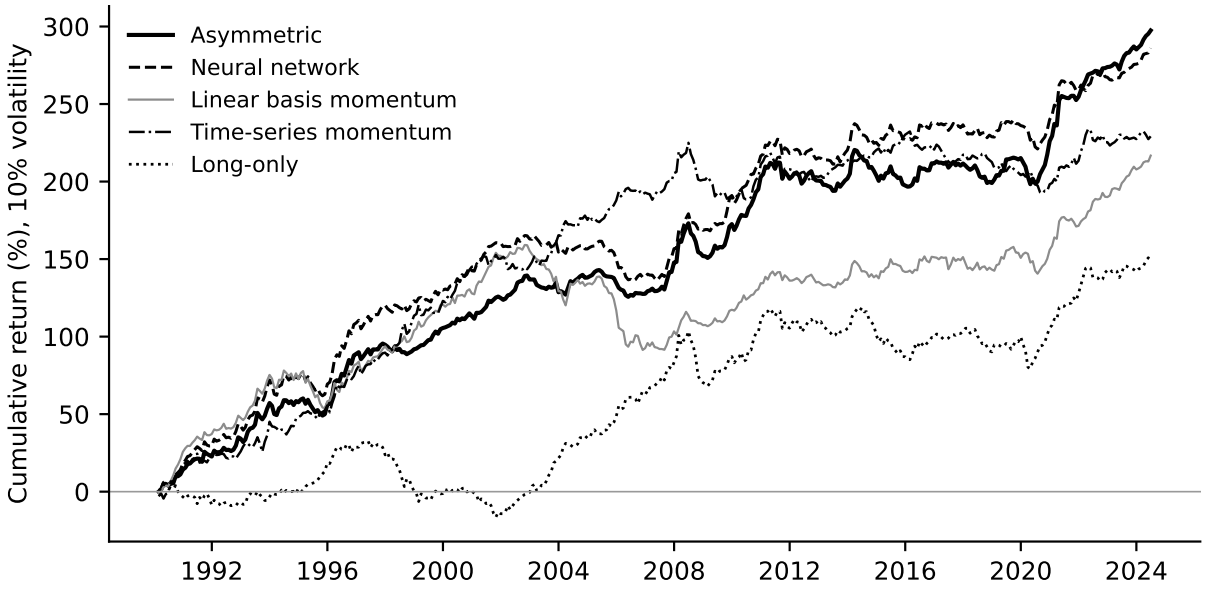


Exhibit A16: **Cumulative Out-of-Sample Returns.** Walk-forward strategies of Exhibit 9 of the article, February 1990–June 2024, each scaled ex post to an annualized volatility of 10%; cumulative sums of monthly returns.

raises the maximum Sharpe ratio even when linear basis momentum is already included (from 1.18 to 1.26). Panel C reports returns in stress periods at 10% volatility. Because the asymmetric position is net long, it loses in commodity crashes, between 1.0% and 4.2% per month, though less than the long-only portfolio.

Exhibit A17: **Implications for Investors.**

Panel A: Long-only tilt, $w_{i,t} = \max(0, 1 + f(s_{i,t}))$

	Sharpe ratio	Active return (% p.m.)	Tracking error (% p.a.)	Information ratio	α on benchmark
Linear BM	0.76	1.30	17.0	0.92	0.99*** (4.03)
Asymmetric	0.76	2.11	28.7	0.88	1.40*** (3.94)
Neural net	0.79	1.76	22.9	0.92	1.30*** (4.28)
Benchmark (long-only)	0.43				

Panel B: Maximum Sharpe ratio of a commodity multifactor portfolio

	Without linear BM		With linear BM	
	Factors	+ Position	Factors	+ Position
Asymmetric	1.16	1.22	1.18	1.26
Neural net	1.16	1.20	1.18	1.20

Panel C: Stress periods, % per month at 10% volatility

	Months	Linear BM	Asymmetric	Long-only	TSMOM
Commodity crash 2008–09	8	−0.87	−2.80	−4.43	−2.72
Oil price collapse 2014–16	19	−0.01	−1.01	−1.75	0.82
COVID-19, 2020	4	−0.75	−4.22	−5.14	0.61
NBER recessions	36	1.09	0.58	−0.82	0.08
Worst 10% long-only months	42	0.57	−2.33	−4.94	0.00
All months	413	0.52	0.72	0.36	0.55
Corr. with long-only, worst 10% / other		0.19 / −0.02	0.59 / 0.50		

Walk-forward out-of-sample positions of Exhibit 9, February 1990–June 2024; $f(s)$ is linear basis momentum, the asymmetric position, or the neural-network position. Panel A: the tilt over- or underweights each market of the equal-weighted, volatility-scaled long-only portfolio by $f(s)$ and never shorts; active return and tracking error relative to the long-only benchmark; information ratio annualized; α in % per month on the benchmark with Newey–West t-statistic. Panel B: annualized maximum (tangency) Sharpe ratio of the long-only portfolio, time-series momentum, and the cross-sectional basis-momentum, momentum, and carry factors (tercile sorts on BM, 12-1 month momentum, and the basis), without and with linear basis momentum, before and after adding the position. Panel C: each strategy is scaled ex post to an annualized volatility of 10%; commodity crash: July 2008–February 2009; oil price collapse: July 2014–January 2016; COVID-19: January–April 2020. ***, **, * denote significance at the 1%, 5%, and 10% levels.

E. The Neural Network

This section supports Section 6.4 of the article. It documents the specification, the selected hyperparameters, the predictive accuracy, and how the estimated shape depends on the design choices.

Exhibit A18 reports the specification. The network has a single input, the signal, and its target is the risk-adjusted return net of the market's mean in the training window, so that it learns the shape of the conditional expectation rather than the average return. The configuration is selected among 32 candidates by the mean squared error on the last four years of each training window, and the predictions average ten networks with different random seeds. Exhibit A19 reports the configuration selected in each estimation window; the higher learning rate is selected in every window, whereas the regularization and the size of the network vary.

Exhibit A18: **Neural-Network Specification.**

Component	Choice
Data	
Input	Signal $s_{i,t}$, clipped at the 1st and 99th percentiles of the training window
Target	$RaR_{i,t+1}$ minus the market's mean in the training window
Training sample	Pooled panel of all markets, all months up to the estimation date
Architecture	
Hidden layers	1 or 2, each with 4 or 8 nodes, tanh activation
Output	1 node, linear activation; biases in all layers
Estimation	
Loss	Mean squared error, $\frac{1}{N}\sum(RaR_{i,t+1} - \bar{r}_i - g(s_{i,t}))^2$
Optimizer	Nadam (Dozat, 2016) ($\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-6}$), batch size 1,024, at most 50 epochs
Learning rate	0.01 or 0.001, cosine decay (Loshchilov and Hutter, 2017)
Regularization	L_2 penalty 10^{-5} or 10^{-4}; dropout 0 or 0.25; early stopping on the validation loss without the L_2 penalty (patience 25 epochs); weights of the best epoch restored
Model selection	
Validation sample	Last four years of each training window
Grid	32 configurations (learning rate × L_2 × dropout × nodes × layers), lowest validation mean squared error (without L_2 penalty)
Ensemble	Selected configuration re-estimated with ten random seeds; mean prediction
Use	
Estimation dates	Full sample (shape); expanding windows with data through December 1989, 1997, 2005, 2013, 2021 (walk-forward) and 1999 (single split)
Position	$f(s_{i,t}) = \hat{g}(s_{i,t})/c$, with c such that f has the root mean square of the (clipped) $s_{i,t}$ in the training window

Specification of the network that estimates $E[RaR_{i,t+1} - \bar{r}_i \mid s_{i,t}] = g(s_{i,t})$ in Section 6.4. Exhibit A19 reports the hyperparameters selected in each estimation window.

Exhibit A19: **Selected Neural-Network Hyperparameters.**

Estimation	Learning rate	L_2	Dropout	Nodes	Layers	Val. loss
Single split (1999)	0.01	10^{-4}	0.25	8	2	1.4980
Walk-forward, data through 1989	0.01	10^{-4}	0	8	2	1.7689
Walk-forward, data through 1997	0.01	10^{-5}	0.25	4	2	1.4147
Walk-forward, data through 2005	0.01	10^{-4}	0	8	2	1.3194
Walk-forward, data through 2013	0.01	10^{-5}	0.25	4	1	1.1235
Walk-forward, data through 2021	0.01	10^{-5}	0	4	2	1.5448
Full sample	0.01	10^{-4}	0	8	2	1.2998

Configuration with the lowest validation mean squared error (without L_2 penalty) in each estimation window (32 candidates, Exhibit A18); the ensemble re-estimates it with ten seeds. Validation loss: mean squared error on the validation sample.

Exhibit A20 compares the forecasts. Panel A reports the out-of-sample R^2 against the prevailing mean of each market. All forecasts beat the prevailing mean (Clark–West test), and the asymmetric regression has the highest R^2 (0.38%), almost twice that of the linear regression (0.21%). The network has the lowest (0.03%), because its flexibility adds estimation noise. Panel B reports the slopes of the network's response for positive and negative signals. The response is steeper for positive signals in every estimation, with the largest differences in the full sample and in the windows through 2005 and 2021, and the smallest

in the window through 1997 and in the single split.

Exhibit A20: **Out-of-Sample Predictive Accuracy and the Shape Learned by the Network.**

	R^2_{OOS} (%)	Clark–West t	
Panel A: Out-of-sample R^2, walk-forward 1990–2024			
Linear	0.206	3.82***	
Asymmetric	0.377	4.95***	
Kernel regression	0.173	3.84***	
Neural network	0.030	2.62***	
Panel B: Slopes of the neural-network response			
Estimation	Slope for $s > 0$	Slope for $s < 0$	Difference
Full sample	1.71	0.22	1.49
Single split (1999)	1.18	0.90	0.28
Walk-forward, data through 1989	1.52	0.78	0.74
Walk-forward, data through 1997	1.02	0.82	0.20
Walk-forward, data through 2005	2.11	0.03	2.08
Walk-forward, data through 2013	1.30	0.91	0.40
Walk-forward, data through 2021	2.18	−0.73	2.90

Panel A: out-of-sample R^2 in percent of forecasts of $RaR_{i,t+1}$ relative to the market's prevailing mean in the training window (Campbell and Thompson, 2008); models re-estimated with data through December 1989, 1997, 2005, 2013, and 2021. Linear and asymmetric: regressions with market fixed effects; kernel: Nadaraya–Watson regression on market-demeaned returns with cross-validated bandwidth; neural network: Exhibit A18. Second column: Clark and West (2007) t-statistic against the prevailing mean, months aggregated, Newey–West standard errors; stars refer to one-sided p-values. Panel B: slopes of a regression of the ensemble-mean position $f(s)$ on $s^+ = \max(s, 0)$ and $s^- = \min(s, 0)$ between the 1st and 99th percentiles of s in the training window, on the position scale; a positive difference means a steeper response to positive signals.

Exhibit A21 and Exhibit A22 change one design choice at a time, and the loss and the intercept together. A longer validation sample and an unclipped input preserve the asymmetry. Symmetrized training data, which impose $g(-s) = -g(s)$, remove it by construction (-0.02), and the Sharpe-ratio loss, which optimizes a position rather than an expectation, nearly removes it (0.11); both variants earn lower Sharpe ratios than linear basis momentum. In every variant, the alpha of the network is insignificant once the asymmetric position is added as a control.

Exhibit A21: **Neural-Network Design Choices.**

Variant	Asymmetry	Sharpe	Δ vs linear	α	α incl. asym.
Main: MSE loss, free intercept, 4-year validation, clipped input	0.74	0.83	0.20*	0.34**	0.01
Validation on the last 8 years	1.90	0.81	0.18*	0.48**	0.06
Input not clipped	0.76	0.84	0.21*	0.37**	0.02
Symmetrized training data, (s, y) and $(-s, -y)$	−0.02	0.56	−0.07	−0.03	−0.22
Sharpe-ratio loss	0.11	0.39	−0.24	0.32	0.16
Intercept forced to zero, $g(0) = 0$	1.17	0.79	0.16	0.26	−0.07
Sharpe-ratio loss and $g(0) = 0$	0.58	0.80	0.17	0.35	0.00

Variants of the network of Exhibit A18, each changing one element of the main specification; the last row combines two. All variants use the same grid of 32 configurations and ten seeds. Asymmetry: median across walk-forward windows of the difference between the slopes of the response for $s > 0$ and $s < 0$ (Exhibit A20, Panel B). Walk-forward February 1990–June 2024: Sharpe ratio of the network strategy, its difference to linear basis momentum (stars: one-sided bootstrap p-value), and its alpha in % per month on linear basis momentum and the long-only portfolio, and additionally on time-series momentum and the asymmetric position (Newey–West t-statistics for stars).

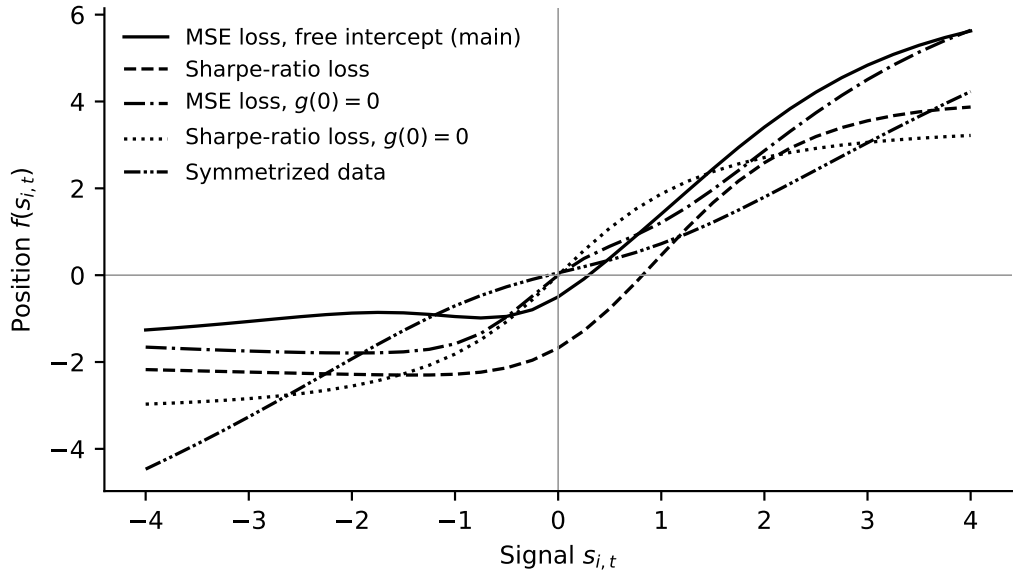


Exhibit A22: **Full-Sample Response of Alternative Network Specifications.** Ensemble-mean response $f(s_{i,t})$ estimated on 1960–2024, between the 1st and 99th percentiles of the signal, for the main specification (mean-squared-error loss, free intercept), the Sharpe-ratio loss, an intercept forced to zero, both restrictions together, and symmetrized training data.

F. Replication

Every number in the article is produced by a script of the replication package, in the following order.

1. Data: Bloomberg roll-adjusted first- and second-nearby prices, with the old NYMEX

gasoline contract rebuilt from unadjusted first- to third-nearby prices on the NYMEX roll calendar and stale prices removed; monthly excess returns; unadjusted prices for the state of the curve; Commitments of Traders positions (CFTC, ICE, LME), last report at least seven days before the month end and at most four weeks old.

2. Signal: basis momentum, Equation (2); signal and risk-adjusted return, Equation (3); 12-1 month momentum.

3. Analyses: linear baseline; model-free shape and asymmetric regressions; regression kink; components of the futures return; state and group tests; out-of-sample evaluation; robustness; neural network (Exhibit A18).

4. Exhibits: one script per table and figure, which reads the stored results and writes the LaTeX file.

The neural network is estimated with fixed random seeds, so that re-running the estimation reproduces all predictions.